

Forecasting long-term US equity returns with a steady-state Bayesian VAR

by

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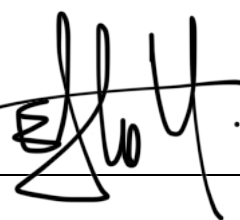
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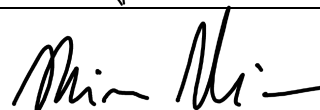
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Abstract

Paper 1. Adapting a Bayesian VAR with steady-state priors first introduced by Villani (2009), and inspired by Shiller et al. (2021) and Lettau et al. (2008), I extend the Ferreira and Santa Clara (2011) framework to allow for reversion of equity valuations to a fair-value based on macroeconomic conditions. The proposed framework generates statistically and economically significant results by: reducing the out-of-sample root mean square forecast error of 10-year US equity returns by annualised 1.2% with respect to the Ferreira and Santa Clara (2011) method and by annualised 2.5% on a model based on simple historical average; and by generating certainty equivalent return gains for a long-term investor of annualised 2% with respect to the Ferreira and Santa Clara (2011) method and of annualised 2.75% on a model based on simple historical average.

Paper 2. I extend Cilla (2023) framework to forecast earnings growth via a Bayesian VAR with steady-state priors. In particular, the newly introduced framework departs from the 20-year moving average assumption of Ferreira and Santa Clara (2011), and it allows for reversion of earnings growth to an equilibrium based on economic conditions. The proposed framework generates statistically and economically significant results by: reducing the out-of-sample root mean square forecast error of 10-year US equity returns by annualised 1.4% with respect to the Ferreira and Santa Clara (2011) method and by annualised 2.7% on a model based on simple historical average; and by generating certainty equivalent return gains for a long-term investor of annualised 2% with respect to the Ferreira and Santa Clara (2011) method and of annualised 2.6% on a model based on simple historical average.

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1 Paper 1: Modelling equity valuations to forecast long-term US equity returns

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1.1 Introduction and literature review

Is it possible to accurately forecast equity returns? If so, should investors use prediction models different than the simple historical average of returns to build their portfolios? These issues have been long debated in the asset pricing literature given the important implications that they have for investors. In fact, whatever the investment horizon and investment style that an investor has, return forecasts are fundamental inputs in order to build portfolios in line with investors' risk profile, preferences and objectives. The literature on forecasting equity returns is vast, however the predictability of returns is still an open question. To understand why, it is important to acknowledge that there are various aspects of the return forecasting problem that the literature has analysed, and these include: 1) the horizon of the forecasts, 2) the definition, estimation and statistical testing of the predictive models, and 3) the mean-reverting feature of predictive variables such as the price-to-earning ratio, especially more recently.

One of the most cited papers in this field of forecasting equity returns is Goyal and Welch (2008) in which the authors perform an extensive review of most of the predictors proposed by the academic literature to forecast monthly and annual returns. They find that “by and large, these models have predicted poorly both in-sample and out-of-sample for 30 years”. They also conclude that “these models would not have helped an investor with access to available information to profitably time the market”. Goyal, Welch, and Zafirov (2023) have recently updated the Goyal and Welch (2008) analyses finding similar results to their previous research and concluding that “much of the extant literature seems obsolete, with a majority

of variables no longer having empirical support even in-sample.” All the models analysed in these papers and a lot of those used in practice by investors are based on the popular so-called predictive regressions, where returns are forecasted by regressing realised returns over lagged predictive variables. However, Nelson and Kim (1993), Cavanagh, Elliott, and Stock (1995), Stambaugh (1999), Lewellen (2004), Torous, Valkanov, and Yan (2004) have questioned the appropriateness of these models on the grounds that the persistence of the forecasting variables and the correlation of their innovations with returns might bias the regression coefficients and affect t-statistics. To avoid these problems, Ferreira and Santa Clara (2011) proposed an alternative approach to predict equity returns, the so called sum-of-the-parts method. Starting from the decomposition of equity returns into dividend yields, changes in price-to-earnings ratio, and earnings growth, the authors show that separately forecasting each of the return components can lead to statistically significant improvements in the forecast accuracy of both monthly and annual equity returns, with respect to a forecasting model simply based on the historical average of returns.

On the other hand, the literature on forecasting long-term returns has shown somewhat more encouraging results, concluding that forecast errors are lower than the ones of short-term return forecasts and that investors that used those models to define their asset allocation would have enjoyed utility gains. The literature includes Fama and French (1988, 2002), Bogle (1991), Arnott and Bernstein (2002), Ibbotson and Chen (2003), Cochrane (2008), the papers in the book “Rethinking the Equity Premium” by Hammond and Leibowitz (2011), Ferreira and Santa Clara (2011)¹, Bogle and Nolan (2015), Siegel (2016), Duarte and Rosa (2015), and Rangvid (2017). It should be noted though that, with respect to short-term forecasts (such as over 1-month or 1-year horizon) testing long-term forecasts (such as over a 10-year horizon) is by definition more challenging due to the more limited availability of non-overlapping data that can be used to test the out-of-sample performance of these models. In fact, out-of-sample tests are often performed on rolling and overlapping 5-year or 10-year data. For this reasons, in

¹They forecast equity returns from a 1-year horizon to a 5-year horizon

order to test the statistical significance of its results, the literature has started to use statistical methods and specifications, such as the ones proposed by Newey and West 1987, and Lazarus, Lewis, Stock and Watson (2018), that take this limitation into account.

In the context of forecasting long-term equity returns, valuation measures such as the Cyclically Adjusted Price Earnings (CAPE) ratio proposed by Campbell and Shiller (1988a, 1998) have been widely used, in particular to gauge over/under-valuation of equity market and to forecast future returns. In fact, as documented in Campbell and Shiller (1988a,b), CAPE values higher (lower) than its long-term historical average have anticipated lower (higher) future 10-year equity returns, building on the intuition that equity valuations are mean reverting over the medium/long-term horizon. However, as a result of the sustained rise of US equity valuations since the recovery post-Global Financial Crisis, market participants and academics² have started to question the validity of the mean reversion assumption of valuations. Similarly, during his keynote speech³ at the NBER Long Term Investing conference in 2019, Campbell noted that the S&P500 earnings yield defined as $1/\text{CAPE}$ ratio has reversed to its long-term historical average only once since late 1990s (Figure 1), potentially behaving more as a random walk process than a stationary one, therefore raising doubts on its mean reverting feature and also on the appropriateness of relying on measures such as the CAPE ratio to forecast equity returns over long-term horizons. However, it is worth noticing that it is not possible to exclude the case of earnings yield being a near-integrated process, that is a variable that shows high persistence and that, however, mean-reverts in long-enough samples as it might be the case of earnings yields. While equity valuations cannot grow indefinitely or go below zero, it is possible that they can stay away from their long-term mean if certain macro-financial conditions occur.

²e.g. see Shiller and Siegel discussion during the conference Financial Markets, Volatility and Crises: A Decade Later organised by Whartons Jacobs Levy Equity Management Center for Quantitative Financial Research. A summary available at <https://knowledge.wharton.upenn.edu/article/siegel-shiller-stock-market/>

³<https://www.nber.org/lecture/ltam-2019-john-campbell-keynote-speaker>

Figure 1: Earnings yields



Dark blue line is the time series of earnings yields defined as the ratio $1/\text{CAPE}$ of the S&P500 index downloaded from Prof. Shiller's database. The light blue line is the expanding-window arithmetic average of earnings yields computed using data from 1927.

In fact, the literature has come up with various reasons that could justify recent raising equity market valuations and lack of (or slower) mean reversion. One possibility is the steady decline of the equity premium in line with Blanchard (1993), Jagannathan, McGrattan, and Scherbina (2000), Fama and French (2002), possibly due to reductions in the costs of stock market participation and portfolio diversification as suggested by Heaton and Lucas (1999), Siegel (1999), Calvet et al. (2004). Lettau, Ludvigson and Wachter (2008) consider an alternative explanation for the declining equity premium and persistently high equity market valuations: a fall in macroeconomic risk, or the volatility of the aggregate economy over time. In fact, lower macroeconomic risk would justify investor demanding lower reward for their investment in equities, therefore pushing equity valuations upward. Moreover, in line with the discount cash flow model, beyond a decline in the equity risk premium, another important factor affecting equity valuations is the real risk-free rate used to discount future expected earnings: as explained in Shiller (2021) an increase in equity valuations is consistent with a reduction in the *real* interest rates, and this is in line with what we have experienced over the

last 30 years.

At the same time, a growing literature in finance has shown that forecasting equity returns can be significantly improved by deviating from static mean-reversion assumption of equity valuations. One approach, exemplified by Lettau and Van Nieuwerburgh (2007) and Pettenuzzo and Timmermann (2011), is to assume that markets suffer structural breaks and thus mean-reversion targets of valuation measures are dependent on some unobservable state of the economy. Other studies include Arnott et al. (2017), Waser (2021), and Drudi and Nucera (2022) that prove that by modelling the relationship of equity valuations and macroeconomic variables such as macroeconomic risk and/or real interest rates can lead to superior equity return forecasts. It is worth noticing though that they all focus on forecasting horizons up to five years and rely on forecasts performed using the so called predictive regressions, based on directly regressing equity returns on lagged values of the predictive variables.

Starting from these findings, the goal of this paper is to extend the Ferreira and Santa Clara (2011) sum-of-the-parts method to allow for valuations that fluctuate around a time-varying steady state. This is achieved by forecasting changes in equity valuations using a steady-state Bayesian vector autoregressive (VAR) model defined by Villani (2009) that, to the best of my knowledge, so far has been mainly used to forecast macroeconomic variables such as inflation. In particular, I extend the literature in two ways. Firstly, inspired by Shiller (2021) and Lettau et al. (2008), I define a fair-value of the US equity earnings yields as a function of real yields and proxies of macroeconomic volatility. I then adapt a steady-state Bayesian VAR model a la Villani (2009) in order to forecast earnings yields over the next ten years, using the estimated fair-value as a prior for the steady-state of the endogenous variable included in the system. In my case, the Bayesian VAR reduces to a Bayesian AR model as, for simplicity, I only use one endogenous variable: earnings yields. Secondly, I use the changes in equity valuations forecasted with the steady-state Bayesian VAR in order to extend the Ferreira and Santa Clara (2011) framework and to ultimately forecast

10-year US equity returns. The proposed framework generates statistically and economically significant results. Firstly, the proposed steady-state Bayesian VAR improves the forecast accuracy of the earning yields with respect to a framework that assumes reversion to long-term historical average of valuations. Secondly, the proposed extension of the Ferreira and Santa Clara (2011) framework reduces the out-of-sample root mean square forecast error of 10-year S&P500 returns by annualised 1.2% with respect to the sum-of-the-parts method, and by annualised 2.5% on a model based on simple historical average. Finally, analysing the case of a long-term investor that invests in a low-cost S&P500 ETF and a US 10-year Treasury ETF, I show that the proposed framework generates certainty equivalent return gains of annualised 2% with respect to the sum-of-the-parts method, and of annualised 2.75% on a model based on simple historical average. To conclude, the proposed framework would represent a valid alternative to forecast long-term US equity returns, while avoiding biases of the predictive regression models and departing from the assumptions of mean-reverting equity valuations to their historical average.

Next sections are organized as follows. In section 2, I provide the details of the methodology used to compute the fair-value of equity valuations. In section 3, I explain how I use the Villani (2009) framework to forecast earnings yields and consequently equity returns. In section 4, I present the empirical results related to the accuracy of return forecasts and the economic gains for long-term investors that would use the proposed method to allocate their capital between US equities and US bonds. In section 5 I share concluding remarks and ideas for future research.

1.2 Methodology - A fair-value of equity valuations

The first important pillar of the proposed method consists in the estimation of the fair-value of US equity valuations over time as a function of its structural drivers. While in principle various valuation measures can be used, I opt for the CAPE ratio proposed by Campbell and Shiller (1988a,b) widely used in the literature of forecasting equity returns. In particular, CAPE is the ratio of current real prices

to 10-year moving average of real earnings and the authors argue that this ratio should have better forecasting power than the current simple price-to-earnings ratio because aggregate corporate earnings display short-run cyclical noise. These characteristics make this measure appropriate for my framework that, ultimately, is used to forecast equity returns.

To understand what the potential drivers of the CAPE ratio are, I start from the discount cash flow model that defines equity prices as future expected companies' earnings discounted at a rate that is function of a risk-free rate plus a risk premium. Therefore, following the Gordon model equity prices can be expressed as:

$$P = \frac{E}{rf + rp - g} \quad (1)$$

where, in the case of the broad US equity market that we consider, P is the price level of the equity index, E are the expected future corporate earnings, rf is the real risk-free rate, rp is the risk premium and g is the expected growth of companies' real earnings. Using the cyclically adjusted earnings for E , as defined in Campbell and Shiller (1988a,b), we can therefore write:

$$\frac{1}{CAPE} = EY = rf + rp - g \quad (2)$$

Therefore according to this framework, earnings yields EY are positively related to the real interest rates and the risk premium, and inversely related to expected growth of companies' per-share real earnings. This is line with the conclusions shared in Lettau et al. (2008) and Shiller et al. (2021).

To perform empirical analyses, for each variable, I define the following proxies justified by approaches and findings documented in the literature. For the real interest rate rf , I use the 10-year real yield measure proposed by Shiller et al. (2021) as the difference between the 10-year nominal yield of US treasuries and the 10-year expected US inflation, where the latter is proxied by the 10-year historical average of realised US inflation. This variable is directly downloaded from Shiller's

database.

Building on the intuition of Lettau et al. (2008), changes in the risk premium rp are proxied by a measure of macroeconomic volatility that, in the base case, is assumed to be the average of the 5-year rolling standard deviations of US real GDP growth rate and the US inflation rate. Although the time window to compute the standard deviation is arbitrary, the 5-year period has been chosen to have enough quarterly data points to compute a precise enough estimator and, at the same time, to produce a variable that accounts for changing medium/long-term economic conditions. As robustness check, I also consider the standard deviation of real GDP and inflation computed over a 3-year and 10-year time window. Both the US real GDP and US inflation series are downloaded from the FRED database of the Federal Reserve Bank of St. Louis. For both variables I only use data starting from 1947.Q1 onwards, given that this is the first available data point for the US real GDP variable.

Given the equity forecasts are generated on a monthly basis and the GDP data is quarterly, I interpolate historical GDP quarterly data points to compute a monthly series by using the methodology proposed by Shiller (2015) to convert the quarterly series of earnings-per-share into a monthly series. As an example, using 1947.Q1 and 1947.Q2 available data points, the monthly data points are computed as a weighted average of the two available data: the April 1947 data point is equal to $2/3$ of March 1947 data point and $1/3$ of the June 1947 data point, the May 1947 data point is equal to $1/3$ of March 1947 data point and $2/3$ of the June 1947 data point. At this stage, it is important to note that all statistics are computed using only available data at any given time, and in particular I lag the GDP data by three months in order to perform a real-time analysis and to avoid look-ahead bias. It is worth noting that I also tested the case of computing the fair-value regression using quarterly data and then using the quarterly fair value for the months in between two consecutive quarters, however this does not lead to significant differences for the analyses that will follow.

For the third component g , defining a proxy is not a straightforward task. Analysts' survey data is notoriously biased and not useful to predict earnings and future returns, see for example Kothari et al. (2016) for an extensive review. At the same time, various researches tend to suggest that changes in equity valuations might be primarily driven by changes in the discount rate, in line with Cochrane (2011). Given these considerations, I therefore opt to exclude the variable from the fair value base case. However, while eventually not used in the final framework, I test the robustness of the fair value regression also including various variables that could proxy earnings expectations. In fact, it is important to see if, including those proxies, the estimated sensitivity parameters of the other variables and/or the overall fit of the regression are affected. Inspired by Shiller et al.'s (2021) approach to compute expected inflation and Ferreira and Santa Clara's (2011) approach to forecast earnings growth, I compute the expected real earning growth rate by using the historical average of per-share real earnings growth. For the time window used to compute the historical average, I use the 10-year moving average. As an additional test, I also use the 10-year moving average of the real GDP growth rate, starting from the assumption that over the long-term real earnings should grow at the same rate of the economy.⁴

To start, I analyse the sample properties of the variables described above, from March 1953 to March 2023. Table 1 reports the results, while Figure 2 displays the time series of the variables used in the main final fair-value framework.

⁴This assumption can be challenging and it will be further analysed in future research. At this stage, it is only used as a robustness check for the sign of the regression parameters estimated in the fair-value model.

Table 1: Sample properties

	EY	R10Y	MVol
Mean	5.77	2.07	0.51
Standard dev.	2.58	1.64	0.25
AR(1)	0.99	0.98	0.99
ADF p-value	0.25	0.08	0.12
EY	1		
R10Y	0.34	1	
Macro Vol.	0.70	-0.11	1

The first two rows of that table report sample means and standard deviations. The AR(1) values are the estimates of an AR model with one lag fitted on each variable. *ADF p-value* numbers are the p-values of the Augmented Dickey-Fuller test for unit root. The lower-triangular matrix reports the pairwise linear correlation coefficients between pair of variables reported in the table. *EY* is the US earnings yield defined as 1/CAPE ratio, *R10Y* is the 10-year US real yield, *MVol* is macroeconomic volatility defined as the annualised average of the 5-year standard deviation of US real GDP growth and US inflation rate. Data used is from March 1953 to March 2023.

Figure 2: Earnings yields, real yields, and macroeconomic risk



The dark blue line is the time series of US earnings yields *EY*. The orange line is the real 10-year yield *R10Y* computed as the difference of the US 10-year treasury yield at time t and the annualised 10-year average of realised US inflation rate up to time t . The gray line is the macroeconomic volatility *MVol* defined as the annualised average of the 5-year standard deviation of US real GDP growth and US inflation rate up to time t . *MVol* is multiplied by 10 in this chart.

Looking at the sign and the magnitude of the correlations among the variables provides already some interesting insights. Firstly, earnings yields are positively

correlated to long-term real yields in line with Shiller et al.'s (2021) intuition and our framework in Equation 1. Moreover, earnings yields are positively correlated with macroeconomic volatility: this confirms the intuition shared in Lettau et al. (2008) that periods with higher macroeconomic volatility are associated with periods of poor equity performance. These results hold also within two subsamples created using 1990 as a cut off date.

These results are confirmed also by analysing the plot of the variables' time series. As we can see from Figure 2, earnings yields tend to fluctuate in parallel with macroeconomic volatility. In particular, it is interesting to see the magnitude of macroeconomic volatility in 1970s and 1980s when both inflation and GDP were particularly volatile, signalling an uncertain environment in which equity valuations were also particularly low (that is, earnings yield was high). While the 1990-2007 era has been characterised by the so called "great moderation", the other significant spikes in macroeconomic volatility were the 2007-09 global financial crisis and, more recently, the 2020 shock. It is interesting to note that while valuations have dropped significantly during the 2007-09 period, the same is true only to a certain extent during the 2020 shock when the US equity market has proven to be relatively resilient, possibly due to accommodative fiscal policies implemented by the US government and low real yields overall.⁵

In order to compute a fair-value of equity valuations and to understand how changes in structural drivers explain movements in the equity valuations, in each month I use a linear multiple regression (Equation 3), estimated with ordinary least squares, to compute the sensitivities of earnings yields to the proposed measures of real yields and macroeconomic volatility, using all the historical available data.

$$EY_t = \alpha_t + \beta_1 R10Y_t + \beta_2 MVol_t + \epsilon_t \quad (3)$$

Before proceeding with the estimation of the fair-value regression, it is

⁵While at this stage I only use a static model, future research could investigate the time-varying nature of the estimated parameters of the fair-value regression.

important to test for stationarity of the variables used and, in case of non-stationarity, for cointegration among them. As reported in Table 1, the augmented Dickey Fuller test on earnings yields suggests that it is not possible to reject the hypothesis of unit root. This is in line with most recent intuition shared by Campbell (2019) of earnings yield behaving more like a random walk, however it is worth noticing that it is not possible to exclude the case of earnings yield being a near-integrated process as discussed earlier. On the other hand, macroeconomic volatility appears to have a unit root according to the test. Therefore, in order to perform correct inference of the estimates of Equation 3 we need to ensure that the non-stationary variables are cointegrated. In this case, following the relationship represented in Equation 2 and the intuition shared in Shiller et al. (2021) and Lettau et al. (2008), there is a strong theoretical grounding on the cointegrating relationship of earnings yield with real yields and macroeconomic volatility. Moreover, on an empirically basis, the Engle and Granger (1987) test shows that the null hypothesis of no cointegration of the nonstationary variables can be rejected at the 95% confidence level.

Table 2 reports the results of the intercept and beta coefficients for each variable included in the fair-value regression.

Table 2: Sensitivity of earnings yields to its macroeconomic drivers

X/Y	EY	EY	EY
Intercept	5.32***	2.09***	0.57***
R10Y	0.20***		0.64***
MVol		6.89***	7.37***
Adj. R^2	0.02	0.50	0.68

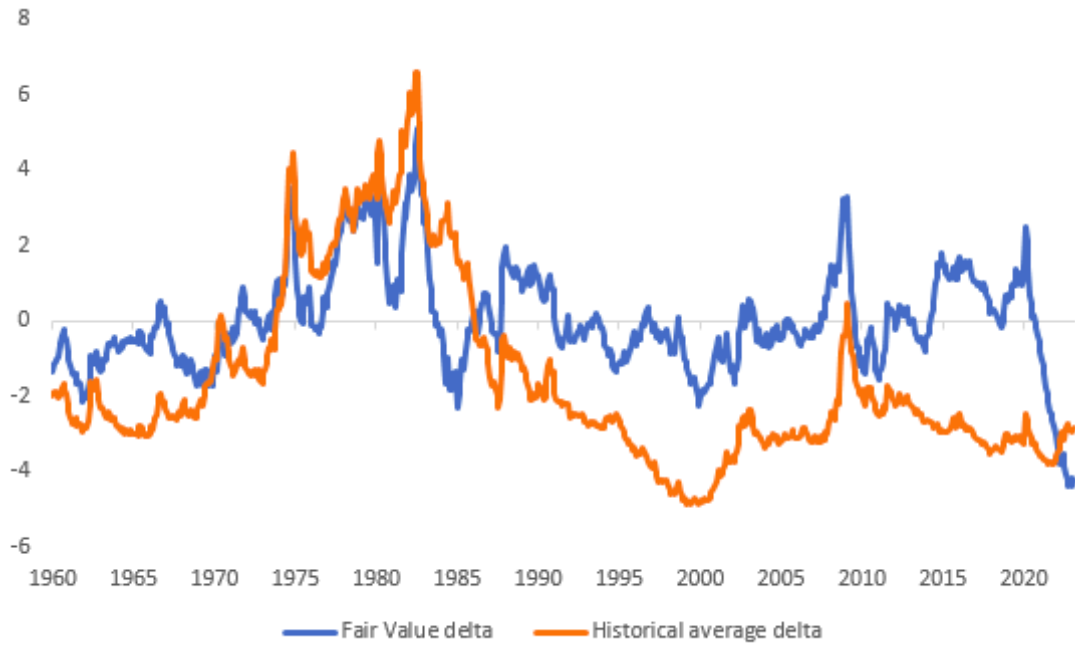
The table reports the coefficient of a linear multiple regression estimated with OLS methods, where earnings yields are regressed on various sets of explanatory variables. $Adj.R^2$ numbers are the adjusted- R^2 statistic for each regression. *EY* is the US earnings yield, *R10Y* is the 10-year US real yield, *MVol* is macroeconomic volatility proxied by the average of 5-year rolling standard deviations of US real GDP growth rate and US inflation rate. Data used is from March 1953 to March 2023. For each regression coefficient, *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels respectively.

Looking at the output reported in Table 2, we can see that the sign of all parameters are in line with the intuition of framework in Equation 1, Shiller et

al. (2021) and Lettau et al. (2006). Focusing on the complete model with real yields and macroeconomic volatility, the sensitivity parameters for real yields and macroeconomic volatility are positive (0.64 and 7.37 respectively) and significantly different from zero. Moreover, including both real yields and macroeconomic volatility as independent variables produces an adjusted R^2 of 68% signalling that movements in the components of the discount rate explain a great amount of the variation of equity valuations over time, in line with the approach followed by Cochrane (2011). Also, comparing the various regression sets shows that the changes in medium/long-term macroeconomic volatility represents the most important factor, explaining alone around 50% of the variation of earnings yields in the sample considered here.

Figure 3 reports the time series of the difference between actual earnings yield and the fair-value (blue line), and the time series of the difference between actual earnings yield and its long term historical average (orange line). Both the fair-value and the long-term historical average have been estimated at each point in time t using all the historical available data up to time t , without look-ahead bias.

Figure 3: Deviations of earnings yields from fair-value and historical average



The blue line is difference between actual earnings yields and the fair-value as defined in Equation 3. The orange line is difference between actual earnings yield and its long term historical average. Both the fair-value and the long-term historical average have been estimated at each point in time using all the historical available data up to March 2023.

When the lines in Figure 3 are above or below the zero line they signal an undervalued or overvalued equity market. In particular, when the blue line is in the positive territory it means that earnings yield is higher than the fair-value estimated according to the model in Equation 3, therefore the US equity market is undervalued according to the fair-value framework. Similarly, when the orange line is in the positive territory, it means that earnings yield is higher than its long-term historical average and therefore the market is potentially undervalued. To start, it is interesting to see that the deviations from the fair value have overall a lower magnitude than the deviations from the historical average. In particular, comparing earnings yield to its historical average would have signalled an overvalued market over the past 40 years or so, apart from few months during the Global Financial Crisis when the earnings yield was in line with the historical average, as noted by Campbell (2019). This is important for return forecasting as this has caused returns forecasts over the past years to be much lower than the actual market performance. In contrast, the fair-value model would suggest that the US equity market has gone through various cycles over the past 60 years with

periods of undervaluation, like for example at the peak of the Global Financial Crisis, followed by periods of stretched valuations, such as during the dot-com bubble in early 2000s, the late-1980s period characterised by high macroeconomic uncertainty, and the more recent 2020 shock. It is interesting to note that in the last 40 years, the 2020 period, characterised by very high levels of volatility of both GDP and inflation, was the only period that would have justified earnings yields to be in line with its long-term historical average, according to the proposed fair-value framework.

To perform robustness checks on the fair-value regression, I now estimate different settings of the fair-value Equation 3 by adding additional factors and changing the definitions of the macroeconomic volatility series. In particular, as explained earlier, I add potential proxies for the expected growth of real earnings: the 10-year moving average of real earnings-per-share growth, and as an additional test, the 10-year moving average of the real GDP growth rate. Results are reported in Table 3.

Table 3: Sensitivity of earnings yields to its macroeconomic drivers

X/Y	EY	EY	EY
Intercept	1.69***	1.69***	0.88***
R10Y	0.59***	0.76***	0.54***
MVol 3Y	6.31***		
MVol		7.49***	7.44***
10Y RGDP		-0.47***	
10Y REPS			-0.03***
Adj. R ²	0.50	0.69	0.68

The table reports the coefficient of a linear multiple regression estimated with OLS methods, where earnings yields are regressed on various sets of explanatory variables. *Adj.R²* numbers are the adjusted-R² statistic for each regression. *EY* is the US earnings yield, *R10Y* is the 10-year US real yield, *MVol* is macroeconomic volatility proxied by the average of 5-year rolling standard deviations of US real GDP growth rate and US inflation rate, *MVol3Y* is macroeconomic volatility proxied by the average of 3-year rolling standard deviations of US real GDP growth rate and US inflation rate, *10YRGDP* is the rolling average of 10-year US real GDP growth, *10YREPS* is the rolling average of 10-year S&P500 real earning-per-share reported earnings. Data used is from March 1953 to March 2023. For each regression coefficient, *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels respectively.

As we can see, the signs of all coefficients are in line with the fair-value identity

of Equation 2. In fact, the sensitivity parameter of $R10Y$ remains always positive. The same is true for the $MVol$ variables in both cases of the average of 3-year and 5-year rolling standard deviations of real GDP and inflation. Finally, when I add proxies for expected growth of earnings, the sign of the sensitivity parameters for those variables are negative in both cases of 10-year real GDP growth and 10-year real earning-per-share growth, as expected.

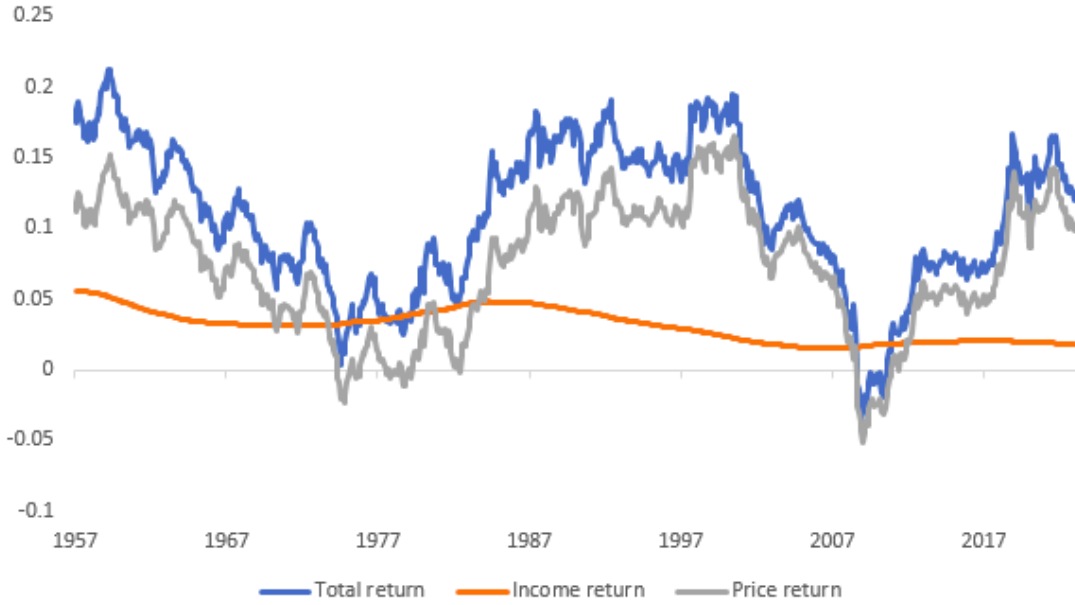
1.3 Methodology - Forecasting equity valuations and equity returns

As reported in the introduction, one of the common framework used in the literature of forecasting returns is the so called predictive regression, where forecasts are generated by regressing realised returns on lagged predictive variables. Despite their simplicity, the estimates of these models are likely to be biased due to the persistence of the predictive variables and the correlation of their innovations with returns. To avoid these problems, Ferreira and Santa Clara (2011) have proposed an alternative method that forecasts returns by forecasting each of the return components separately. Their method is based on the intuition that equity returns (r) can be decomposed into three components: an income return component, namely the dividend-price ratio (dy) in the case of equities, and a price return component that can be further decomposed into earnings growth (Δe) and growth of the price-to-earning ratio (Δpe) - as shown in Equation 4.

$$(1 + r) = (1 + \Delta e) * (1 + \Delta pe) * (1 + dy) \quad (4)$$

If we analyse what has happened historically by decomposing the rolling 10-year total returns of the S&P500 index, it is interesting to notice that the price return component has explained on average around 70% of the annualised returns over time, while the income return component has explained the remaining 30%, as shown in Figure 4. This contributes to justify why a lot of the attention of both academics and practitioners has been on modelling the components of equity price returns.

Figure 4: 10-year total return components: income return and price return



The blue line is the rolling annualised 10-year total return of the S&P500 index computed using data from March 1947 to March 2023. available data up to March 2023.

After decomposing the returns, it is important to accurately forecast each of the components. On the one hand, the income return component and the dividend-price ratio are notoriously hard to predict with accuracy, in line with Cochrane (2008). On the other hand, on the price return side and in particular for changes in valuations, a popular assumption in academia and the industry is that equity valuations revert to their historical average over the medium/long-term: therefore if earnings yields are higher (lower) compared to their historical average it is reasonable to expect higher (lower) returns in the future. However, as we have seen earlier, valuations have not reverted to their long-term mean for around 40 years causing forecasts of equity returns to be lower than the realised returns. The contribution of this paper is to introduce a framework that replaces the assumption of valuations reverting to the historical average with the assumption of valuations reverting to their fair-value based on real yields and macroeconomic volatility, as defined in section 2. The reason for relying on the proposed fair-value is that the speed of the reversion to the equilibrium might be higher or lower based on certain macroeconomic conditions, as suggested by the literature discussed earlier. Therefore the key idea of the proposed framework is that if valuations are above or

below its fair value at a given point in time, it is reasonable to assume a reversion of valuations to the fair value (not their historical average) in the future. As long as the current fair-value of valuations is away from their historical average, changing the convergence level of valuations can lead to significantly different forecasted valuations and consequently different return forecasts between the two methods.

1.3.1 Forecasting equity valuations

In order to model the reversion of valuations to the proposed fair-value estimated in section 2, I use a steady-state Vector Autoregression (VAR) framework as in Villani (2009). Autoregressive models have long been used in the financial literature to forecast equity valuations and returns, e.g. Campbell and Viceira (2002) among others, producing forecasts for the endogenous variables that by definition converge to their long-term historical sample average. However, as Villani shows, departing from the mean-reverting assumption of the VAR forecasts and using prior information on the long-term convergence level of the endogenous variables included in the VAR can potentially lead to substantial gains in forecast accuracy.

Following Villani (2009), the steady-state Bayesian VAR system is defined as follows:

$$\begin{aligned}\Pi(L)(x_t - \Psi) &= \epsilon_t \\ \epsilon_t &\sim N(0, \Sigma)\end{aligned}\tag{5}$$

where x_t is a vector of endogenous variables, Π is a lag polynomial defined as $\Pi(L) = I - \Pi_1 L - \Pi_2 L^2 - \dots - \Pi_p L^p$, p is the number of lags of the VAR. Following Villani (2009), the *random* variables Ψ represent the unconditional mean of the process as $E(x_t) = \Psi$. In our case, for simplicity, I only include earnings yields as endogenous variable in the VAR system, that therefore reduces to a single variable steady-state AR framework. As we can see, the long-term steady state level is treated as a model parameter that needs to be estimated. Given the Bayesian

framework, the estimation starts by defining priors for each parameter, for which I follow Villani (2009).

In particular, the prior on Σ is taken to be:

$$p(\Sigma) \propto |\Sigma|^{-(p+1)/2}$$

The prior for $vec(\Pi)$ is a general multivariate normal distribution:

$$vec(\Pi) \sim N(\Theta_{\Pi}, \Omega_{\Pi})$$

Then, the prior for Ψ is

$$vec(\Psi) \sim N(\Theta_{\Psi}, \Omega_{\Psi})$$

In order to define the parameters of the prior distribution I follow Villani's approach. In particular, the prior proposed by Litterman (1986) is used on the coefficients in Π , with the default values on the hyperparameters in the priors advocated by Doan (1992), therefore the overall tightness is set to 0.2. Moreover, the prior mean on the first own lag in Π is set to 0.5. This is in line with the view that earnings yield should revert to the steady-state however with a convergence speed that is moderate: 0.5 is in fact half way between a prior close to 0, that would indicate a very fast convergence, and a prior close to 1, that would indicate a very slow convergence. In my case, only the prior on the first own lag is needed as p is set equal to 1: this is justified by the fact that the first own lag already explains most of the variation of the earnings yield over time.

Next, in order to estimate the posteriors, it is important to notice that the posterior distribution of the VAR system defined above is intractable. However, Villani (2009) shows that the posterior distribution of each set of model parameters conditional on the other parameters is tractable, and that a Gibbs sampler can be used to draw from the joint posterior. In particular, the conditional posterior of

Σ is an inverted Wishart distribution:

$$\Sigma|\Pi, \Psi, D \sim IW(E'E, T)$$

where $E = \Pi(L)(x_t - \Psi d_t)$.

The posterior for $vec(\Pi)$ is:

$$vec(\Pi)|\Sigma, \Psi, D \sim N(\bar{\Theta}_\Pi, \bar{\Omega}_\Pi)$$

where $\bar{\Omega}_\Pi^{-1} = \Sigma^{-1} \otimes X'_\Psi X_\Psi + \Omega_\Pi^{-1}$, $\bar{\Theta}_\Pi = \bar{\Omega}_\Pi[vec(X'_\Psi Y_\Psi \Sigma^{-1}) + \Omega_\Pi^{-1} \Theta_\Pi]$, $Y_\Psi = [x_t - \Psi d_t]$, $X_\Psi = [x_{t-1} - \Psi d_{t-1}, \dots, x_{t-k} - \Psi d_{t-k}]$.

Then, the posterior for Ψ is

$$vec(\Psi)|\Pi, \Sigma, D \sim N(\bar{\Theta}_\Psi, \bar{\Omega}_\Psi)$$

where $\bar{\Omega}_\Psi^{-1} = U'(D'D \otimes \Sigma^{-1})U + \Omega_\Psi^{-1}$, $\bar{\Theta}_\Psi = \bar{\Omega}_\Psi[U'vec(\Sigma^{-1}Y'D) + \Omega_\Psi^{-1} \Theta_\Psi]$, $Y = [\Pi(L)x_t]$, $U' = (I_p q, I_q \otimes \Pi'_1, \dots, I_q \otimes \Pi'_k)$, $D = [d_t, -d_{t-1}, \dots, -d_{t-k}]$.

The posterior distribution of the steady state parameter is estimated by combining information included in the data and the prior information. For the steady-state variable in our framework, the prior comes from the fair-value model estimated in section 2. As in any Bayesian framework, the posterior distribution of the parameter will converge to the prior distribution as the uncertainty of the prior reduces. On the contrary, a wide distribution of the prior will lead the estimate of the posterior to be close to the information contained in the historical data. In my case, a tight steady state prior will cause the long-term forecast of earnings yields to be close to the estimated fair value, while a wider prior will yield forecasts of earnings yields that will converge to the long-term historical average. It is important to stress that for the steady-state VAR, convergence to the target distribution is in general fast unless the prior on the steady state is very weakly informative, in which case the steady state VAR should not have been used

in the first place. Given the goal of this paper is to assess whether information from the fair-value model can improve earnings yield forecasts, I assume a tight distribution around the steady-state prior. In specific, at each time t , I use the fair-value estimate at time t to set the mean of the prior distribution and, to make the prior informative, I define the variance of the distribution by assuming a 95% probability for the earnings yield to be in the range of $\pm 0.5\%$ around the fair value. As an example, if at a given time the fair-value regression suggests a level of earning yields equal to 4%, the steady-state prior is assumed to be Normally distributed with a mean of 4% and variance such that there is a 95% probability of being between 3.5% and 4.5%, where the variance is simply the sample variance estimated using all available historical data.

1.3.2 Forecasting equity returns

In order to generate equity return forecasts consistent with valuations forecasted from the steady-state VAR, I extend the Ferreira and Santa Clara (2011) 'sum of the parts' framework. Additionally, it is important to note that Δpe and dy are linked through the payout-ratio (d/e) defined as the ratio of dividends over earnings. Therefore, starting from the decomposition of equity returns into dividend-price ratio (dy), earnings growth (Δe) and growth of the price-to-earning ratio (Δpe), we get:

$$\begin{aligned}(1 + r) &= (1 + \Delta e) * (1 + \Delta pe) * (1 + dy) \\ &= (1 + \Delta e) * (1 + \Delta pe) * \left(1 + \frac{d}{e} \frac{1}{pe}\right)\end{aligned}\tag{6}$$

From Equation 6 we can conclude that the forecast of US equity returns can be computed by forecasting earnings growth, changes in price-to-earning ratio and the payout-ratio. For each components different assumptions can be made and I therefore test various settings defined as follows.

- *FV-VAR2*. In order to forecast earnings growth, I use the 20-year moving average of reported earnings-per-share of S&P 500 companies, in line with Ferreira and Santa Clara (2011). For changes in price-to-earning ratio I

use the forecasts of changes in CAPE ratio produced with the steady state VAR, while for the pay-out ratio I simply compute the full sample historical average.

- *FV-VAR1*. In order to analyse how assumptions on the dividend-price component affect the results of the steady state VAR method, FV-VAR1 mirrors FV-VAR2 apart from the assumption on dividend-price ratio that is now assumed to follow a random walk, therefore forecasting dy by using the most recent available data at each point in time.

Along with the newly introduced FV-VAR frameworks, I also forecast equity returns using alternative models popular in the literature and among practitioners.

- *Hist. Avg.* The first alternative model is the historical average of realised returns computed as the geometric annualised average of realised return up to the time the forecast is generated:

$$r_{t+120} = \prod_{i=1}^t (1 + r_i)^{12/t} - 1$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, r_i are the monthly total returns of the S&P500 index at time i .

- *CAPE*. The second alternative model is based on the predictive regression defined as follows:

$$r_{t+120} = \alpha + \beta * CAPE_t + \epsilon$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, $CAPE_t$ is the Shiller's CAPE ratio at time t . The parameters α , β and ϵ are estimated using an expanding window OLS regression, with the time-series of rolling 10-year annualised returns as dependent variable and the time series of the Shiller's CAPE ratio lagged by 10 years as independent variable.

- *EY*. The third alternative model is similar to the CAPE method but replacing the CAPE ratio with the earnings yield ($1/\text{CAPE}$), computed as follows:

$$r_{t+120} = \alpha + \beta * EY_t + \epsilon$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, EY_t is the inverse of the Shiller's CAPE ratio at time t . The parameters α , β and ϵ are estimated using an expanding window OLS regression, with the time-series of rolling 10-year annualised returns as dependent variable and the time series of the inverse of Shiller's CAPE ratio legged by 10 years as independent variable.

- *FSC2011*. The fourth alternative model is the base case of the ones proposed by Ferreira and Santa Clara (2011), computed as follows:

$$(1 + r_{t+120}) = (1 + \Delta e_{t:t-240}) * (1 + dy_t)$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, $\Delta e_{t:t-240}$ is the 20-year moving average up to time t of reported earnings-per-share growth of the S&P 500 companies, dy_t is the dividend yield at time t (therefore assuming the dy follows a random walk process).

- *VAR2*. The fifth alternative model is similar to the FV-VAR2 frameworks with the difference that growth of price-to-earnings is forecasted using a frequentist VAR with no steady-state priors, whose forecasts therefore converge to the historical average of the endogenous variable.
- *VAR1*. VAR1 mirrors VAR2 apart from the assumption on dividend-price ratio that is now assumed to follow a random walk, therefore forecasting dy by using the most recent available data at each point in time.

It is important to stress that, contrary to the CAPE and EY methods, the FSC2011, VAR and FV-VAR methods do not rely on the so called predictive

regression approach.

1.4 Empirical analyses - Forecast accuracy and investors economic gains

In order to assess the value added of the proposed framework, I compare the forecast accuracy of the proposed and alternative models in forecasting out-of-sample US equities returns. Moreover, I also compute the economic gain of an investor that would use each of those methods to build a buy-and-hold portfolio made of US equities and US bonds with a ten-year investment horizon.

1.4.1 Data

Beyond the data used to compute the fair value of equity valuations presented in sections 2, all data used to perform empirical analyses is downloaded from Shiller database: earnings-per-share, dividend-per-share, price of the S&P500 index, and 10-year real yield⁶. For all methods that use earnings, I lag earnings data by six months in order to make the analysis real-time. This also applies to the CAPE ratio that therefore is a slightly different version of the one available in Shiller database.

The full sample data used to perform empirical analyses goes from March 1947 to March 2023, with the first forecast generated in January 1970 for the 10-year period going from February 1970 to January 1980. In fact, the first 10 years of the sample are used to estimate the first fair-value data point, and 10 years are also used as estimation window for the methods that use a VAR framework. Moreover, all model performance evaluation analyses are generated using all the available data, as well as for the sub-periods with data before December 1990 and with data after December 1990, in order to analyse any potential differences in results over time. The 1990 cut off date has been chosen as it represents the

⁶www.econ.yale.edu/shiller

period in which earnings yields start to be characterised by a slower reversion to its historical average.

1.4.2 Forecast evaluation measures: RMSFE and CER

In order to assess the accuracy of each model's forecasts, I compute the out-of-sample root mean square forecast error (RMSFE), defined as:

$$RMSFE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where y_i are the realised values of the forecasted variable, $\hat{y}_i$ are the forecasts computed with the methods presented earlier, and n is the number of forecasts generated over the sample period.

Beyond forecast accuracy, it is important to test if methods can generate positive economic gains for investors that use them to define their asset allocation. Kandel and Stambaugh (1996), Cenesizoglu and Timmermann (2012) and Timmermann (2018) argue and present evidence suggesting that weak forecasting models can produce economic gains. Cederburg, Johnson and ODoherty (2019) argue that the opposite is also true: good forecasting models used in asset allocation might not necessarily produce economic benefits. In order to analyse economic gains, I first compute the optimal portfolio made of US equities (approximated by the S&P500 index) and 10-year US Treasury bonds, assuming the investor solves the following optimisation problem in each month t :

$$w_t = \arg \min_w \mathbb{E}_t [U(W_{t+h})]$$

$$U(W_{t+h}) = \frac{W_{t+h}^{1-\gamma}}{1-\gamma}$$

$$W_{t+h} = \prod_{i=t}^{t+h} [1 + w_t * r_i^e + (1 - w_t) * r_i^b]$$

where γ is the investor risk aversion (with $\gamma \geq 0$ and $\gamma \neq 0$), W_{t+h} is the total cumulative wealth generated over h months starting from time t , h is equal to 120

(10-year of monthly data), w_t is the optimal portfolio weight to risky asset that is assumed to remain constant over the investment period, r_i^e is the realised equity return at time i , and r_i^b is the realised bond return at time i . The portfolio weights w are therefore static weights, meaning the investor rebalances the portfolio to the strategic benchmark every month until year 10. In line with Illmanen (2011) among others, I forecast future 10-year bond returns using the current 10-year nominal US Treasury yield available at the beginning of each 10-year investment period.

Subsequently, following Sakkas and Tessaromatis (2022), I analyse the economic performance of asset allocation strategies using a prediction model computing the average out-of-sample realized utility of terminal wealth of 520 asset allocation strategies defined in each month. In particular, I calculate the certainty equivalent return (CER) realized by the investor defined as:

$$CER = [(1 - \gamma)\bar{U}]^{\frac{1}{1-\gamma}} - 1$$

$$\bar{U} = \frac{1}{T_{oos}} \sum_{i=1}^{T_{oos}} \frac{W_{i+h}^{1-\gamma}}{1-\gamma}$$

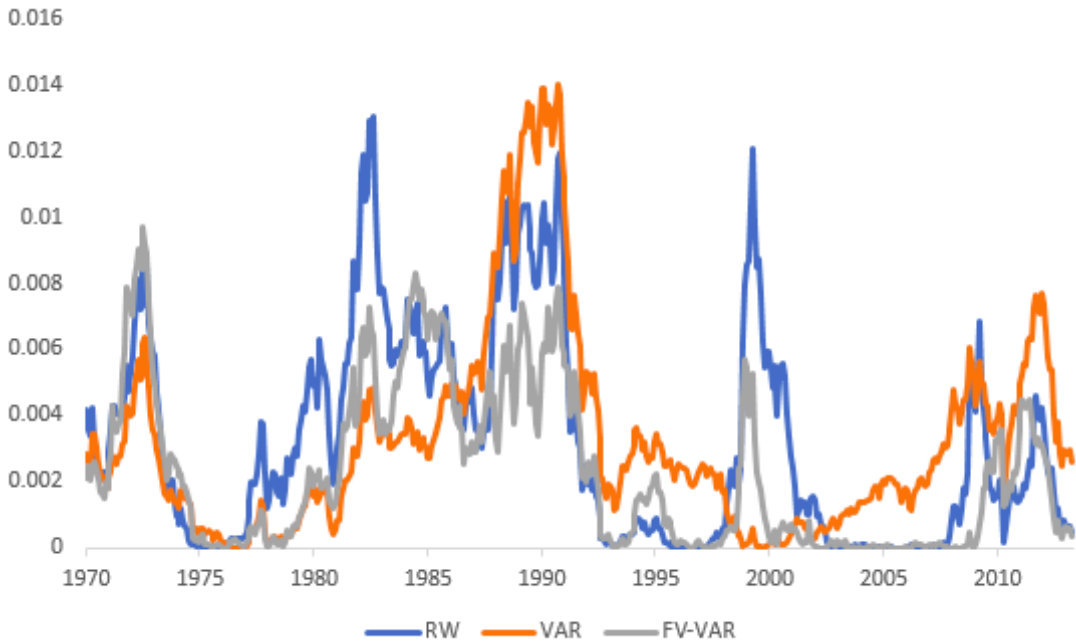
where γ is the investor risk aversion, h is equal to 120, T_{oos} is the total number of months used to compute out-of-sample statistics (that in this analysis is equal to 520), W_{t+h} is the total cumulative wealth generated over h months starting from time t .

In order to analyse the economic gains for different types of investors, I consider three cases. In the first case, I follow Ferreira and Santa Clara (2011) in setting the risk aversion γ equal to 2. In the second case, in order to create portfolios without short positions or leverage, I set the risk aversion equal to 2 and I constrain the risky asset weight w to be between 0% and 100%. In the third case, I set the risk aversion equal to 4 without constraints on w .

1.4.3 Forecast accuracy: equity valuations

To start, Figure 3 reports the time series of the out-of-sample squared forecast error of the 10-year annualised percentage changes in equity valuations, where valuations are forecasted using various methods: the steady-state fair-value VAR (gray line), the random walk forecast (blue line), and a standard frequentist VAR (orange line).

Figure 5: Squared forecast errors of 10-year % changes in valuations



All lines are squared forecast errors (SFE) of 10-year % changes in valuations computed with different methods. The blue line is the SFE of forecasts generated with a random walk assumption, that is the forecasted values are equal to the most recent available value. The orange line is the SFE generated from a frequentist VAR model. The gray line is the SFE of the FV-VAR model.

Overall, the FV-VAR generates most accurate forecasts, however as we can see from Figure 3 the model is not always the best at any point in time, but it is also never the worst. In particular, for the forecasts generated in the late 1990s, when valuations reached a peak during the so called dot-com bubble, a random walk model generates the least precise forecasts, while the frequentist VAR the most precise ones since it is based on the mean-reverting assumption of valuations to their historical average. Apart from this period, in the post-1990 era, the FV-VAR performed the best thanks to an estimated fair-value of earning yields that is

significantly lower than their long-term historical average. In fact, the frequentist VAR is the worst performer in the same time period. This is a sign that adjusting the long-term convergence of the VAR forecasts based on macroeconomic and financial conditions helps to improve forecasts of earnings yields over a 10-year horizon.

1.4.4 Forecast accuracy: equity returns

Next, it is important to test if improvements in forecast accuracy of earnings yield generates more precise 10-year total return forecasts. Table 3 reports the out-of-sample RMSFE for the 10-year annualised total return of the S&P500 index using the methods described earlier. In order to test whether RMSFE of the various methods are statistically different from the RMSFE of historical average forecasts, I use a Diebold Mariano test with Newey and West (1987) standard errors with the lag truncation parameter value based on Lazarus, Lewis, Stock and Watson (2018).

Table 4: RMSFE of forecasting models

	Hist. avg.	CAPE	EY	FSC2011
All	5.9	5.24	4.75*	4.65**
pre-1990	5.7	2.51***	3.53***	3.55**
post-1990	5.96	5.89	5.09	4.97

	VAR1	VAR2	FV-VAR1	FV-VAR2
All	4.8	4.1*	3.58***	3.49***
pre-1990	2.05***	1.85***	2.44***	2.15***
post-1990	5.43	4.63	3.9***	3.84***

The table reports the RMSFE of each forecasting method. *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/CAPE$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields and macroeconomic volatility estimated with a steady-state Bayesian VAR. Data used to estimate the models and to perform out-of-sample forecast evaluation is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

Firstly, the performance of the historical average forecast is the worst among all methods with an annualised average error of 5.9% over the entire period analysed, 5.7% over the sample before 1990, and 5.96% in the sample after 1990. Moreover, over the entire period, both the CAPE and EY models perform better than the historical average method, however only the EY method generates results that are statistically different at 10% confidence level, in line with Siegel (2016). CAPE and EY regressions perform significantly better especially in the pre-1990 sample when valuations were characterised by a stronger mean reversion compared to the post-1990 period. In the sample period post-1990 the CAPE method generates forecast errors that are in line with the historical average benchmark, while the EY method has a lower RMSFE than the historical average method (by about 1%) but not statistically different. The better performance of the EY method with respect to the CAPE method can be explained by the fact that earnings yields, computed as $1/CAPE$, have a zero lower bound that tends to reduce the impact of larger values of the CAPE ratio.

Next, over the entire period, the FSC2011 method performs better than the historical average, the CAPE and EY methods, with gains that go from 10 to 130 basis points. Also, in the pre-1990 and post-1990 periods, the FSC2011 performs better than the historical average method, however differences are significantly

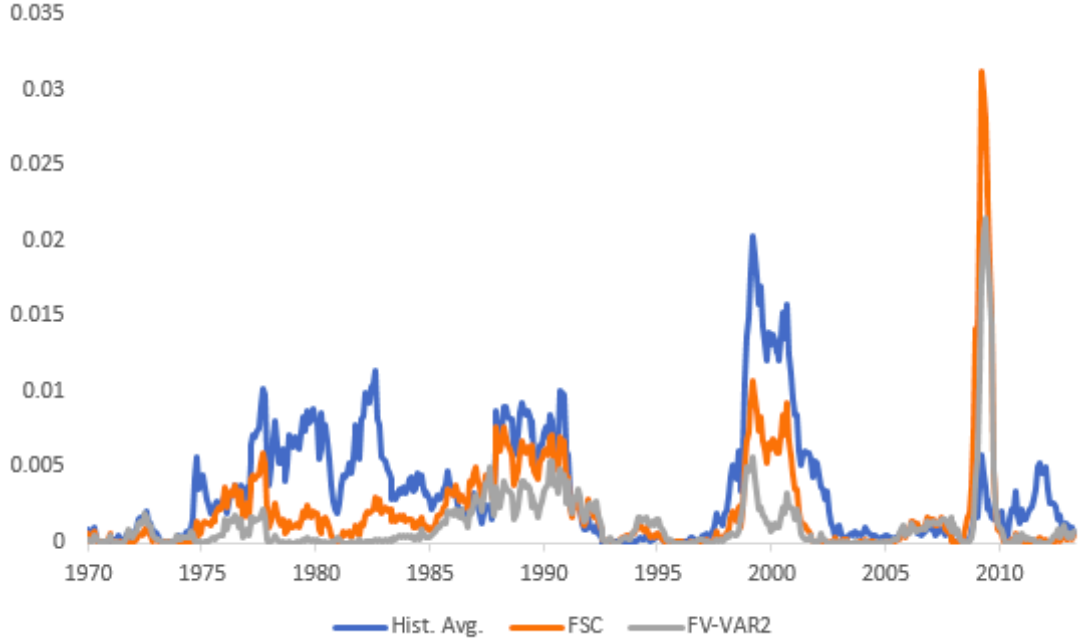
different only in the pre-1990 period: this could be explained by the higher earnings volatility experienced in the post-1990 period. Also, its performance is worse than CAPE method in the pre-1990 period, justified by the fact that FSC2011 does not take into account the mean reverting feature of valuations.

Moving on to the VAR1 and VAR2 methods, over the entire period, they perform better than the historical average, CAPE, EY methods, while only VAR2 performs better than the FSC2011. In particular, VAR2 reduces the RMSFE by around 1.8% annualised with respect to the historical average (statistically different at 10% confidence level) and almost 60 bps to the FSC2011. Taking into account the mean reverting feature of valuations improves the results for both VAR1 and VAR2 compared to FSC2011 in the pre-1990 as expected, however only the VAR2 model performs better than both EY and FSC2011 methods in the post-1990 periods. These results prove that including information about mean reversion of valuations is important overall, however in the post-1990 period, VAR2 performs better than other previously analysed methods as it benefits from an offsetting effect of forecast errors of the dividend yield component and the valuation growth component. In fact, given the structural connection between dividend yield and valuation growth expressed in Equation 6, a higher error in dividend yield forecasts is directly translated into a lower error in the forecasts of valuation changes. The degree of this offsetting effect is determined by the payout ratio: the higher payout ratio the higher the offsetting effect between the errors of the two components.

Interestingly, the fair value VAR models are the best performers in terms of RMSFE over the entire sample and in particular in the post-1990, with no significant difference between the FV-VAR1 and FV-VAR2 approaches. In fact, for the both FV-VAR1 and FV-VAR2, the RMSFE is almost 2.5% lower than the historical average method, 1.7% lower than the CAPE method, 1.1% lower than the EY and FSC2011 methods, 1.2% lower than VAR1, and around 70 basis points lower than VAR2. This is the results of a stronger performance in both sub-samples but in particular in the post-1990 period where both FV-VAR1 and FV-VAR2 are the only methods that produce RMSFEs that are statistically

different from the one generated using the simple historical average approach. These results corroborate and extend the findings of Lettau and Van Nieuwerburgh (2006) that taking into account changing market and economic conditions over time significantly improves long-term US equity returns forecasts.

Figure 6: Squared forecast errors of 10-year US equity returns



All lines are squared forecast errors (SFE) of US earnings yields computed with different methods. Blue line is the SFE of forecasts generated with a random walk assumption, that is the forecasted values are equal to the most recent available value. of the FV-VAR model, orange line is based on forecasts of the VAR model, gray line is based on forecasts of a random-walk model.

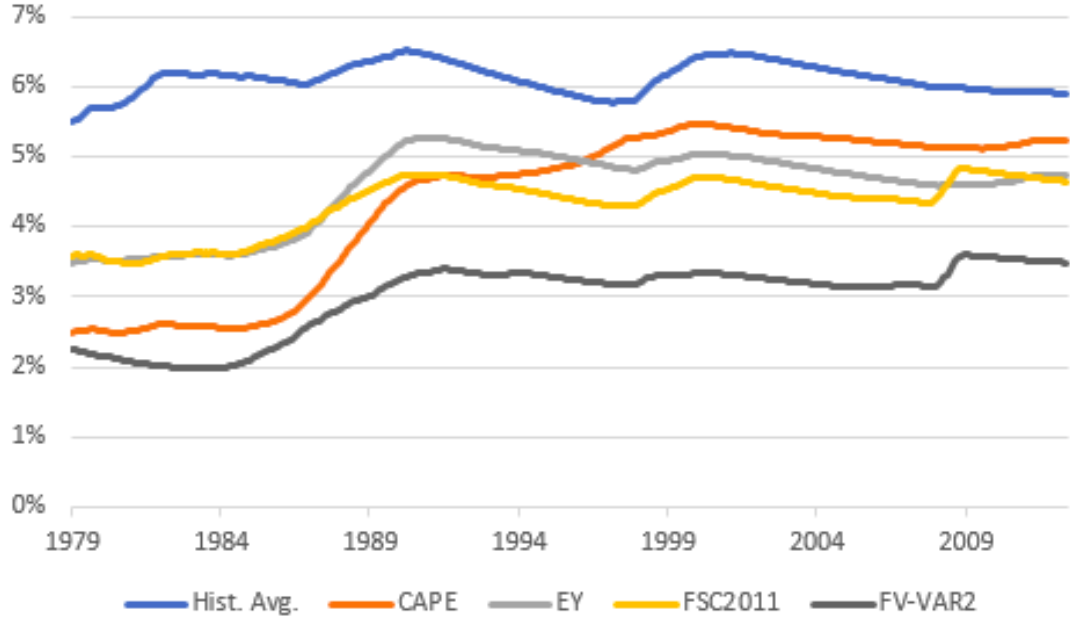
Figure 6 reports the time series of the squared forecast errors that are used to compute the RMSFE numbers reported in Table 4. This confirms that the historical average method generates the highest amount of forecasting errors practically at any time analysed here, apart from the forecasts generated during the Global Financial Crisis. This was a time characterised by a very high volatility of earnings which leads to extreme equity return forecasts due to large swings of the 20-year earnings growth averages that is used to forecast future earnings growth in both the FSC and FV-VAR2 methods.⁷ At the same time, FV-VAR2 method performs best at almost all the periods analysed, in particular the forecasts produced from 1975 to 1990, in late 1990s and early 2000s, and in periods after

⁷Volatility in earnings growth will be the focus of future research

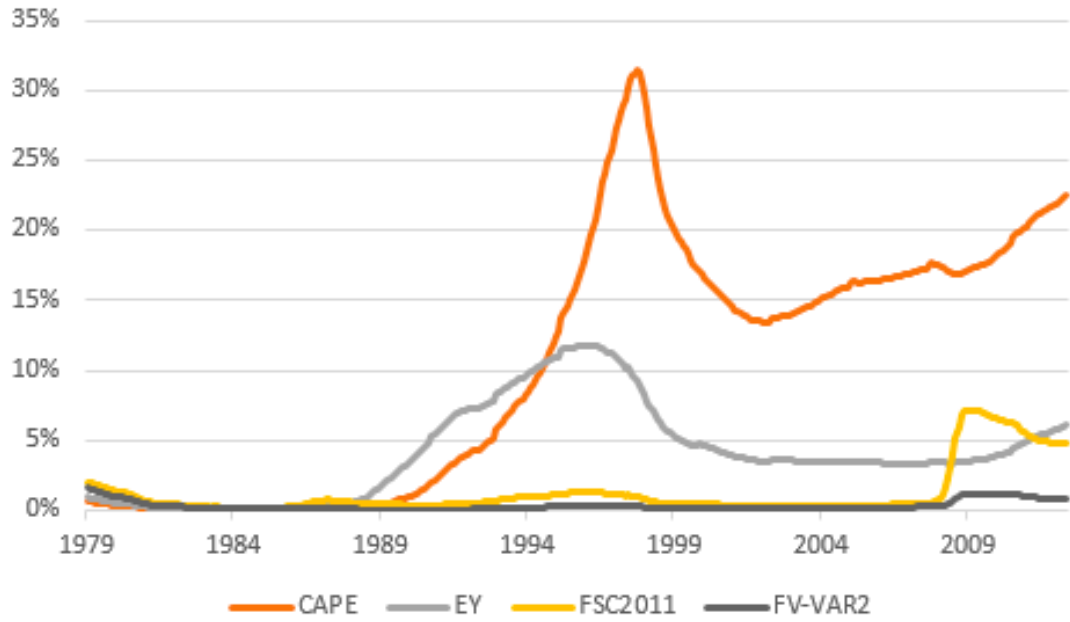
2010. It is interesting to note that while FV-VAR2 forecasts perform poorly during the Global Financial Crisis as explained earlier, its errors are lower than the FSC method thanks to the fact that equity valuations were particularly cheap during those times and this helped offset the drop in the 20-year earnings growth average used to forecast future earnings growth in the sum-of-the-parts approach.

As an additional analysis, inspired by Goyal and Welch (2008), Figure 7 reports the average RMSFE over rolling expanding windows, with a 10-year starting window. Therefore, the last data point of the lines reported in Figure 7 are the RMSFEs (and their p-values) reported in Table 3 for the total sample case. While the historical average method is the worst across all time periods analysed, the FV-VAR2 performs always best among the alternative tested. In line with results reported in Table 3, the CAPE model ranks second until early 1990s, before reaching RMSFEs close to 5% and beyond. On the other hand, the FV-VAR2 method produces a RMSFE close to 2% in the first periods, and then RMSFEs just above 3% as more recent data is used.

Figure 7: Rolling RMSFEs and their p-values



(a) Rolling RMSFE of forecasting models



(b) Rolling pvalues of RMSFE of forecasting models

Lines in subfigure 'a' are the RMSFE of various forecasting models, computed with rolling expanding windows with the first window equal to 10 years. Lines in subfigure 'b' are the p-values of the Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, computed with rolling expanding windows with the first window equal to 10 years. *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *FV-VAR2* is the version of the FSC2011 with valuations reverting to a fair-value based on real yields and macroeconomic volatility estimated with a steady-state Bayesian VAR. X-axis reports the dates in which the forecast for the following 10-year equity returns is produced.

Figure 7b reports the p-values statistics of the RMSFEs shown in Figure 7a. In particular, these are the p-values of the Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018). Combining the results of Figure 7a and Figure 7b, we can conclude that the FV-VAR2 method produces RMSFEs that are always statistically significantly lower than the one produced by the simple historical average method. On the contrary, the CAPE method produces RMSFEs that while lower than the ones of the historical average method, are not statistically different once post-1990 data is used. Interestingly, FSC2011 method's forecasting errors are always significantly lower than the errors of the historical average method for all forecasts generated before 2009 data is used, while p-values jump to around 5% after that period: as explained earlier, volatility in earning during the Global Financial Crisis contributes to generate equity return forecasts that are less accurate.

Lastly, as a robustness check, in Table 5 I report the results of the FV-VAR1 and FV-VAR2 computed using the fair-value model with the macroeconomic volatility defined as the average of the 3-year rolling standard deviations of US real GDP growth and US inflation, instead of 5-year standard deviations used in the base case. As we can see, results are in the line with the ones generated using the base case fair-value setting.

1.4.5 Economic gains

Next, given the improvements in the forecasting accuracy of FV-VAR model, it is now important to assess the economic benefit for an investor that would use each method to build a portfolio with a 10-year investment horizon. Table 4 reports the results in terms of CER gains of each forecasting method (in columns) compared to the historical average method, and for each type of investor (in rows). It is worth mentioning that I do not consider transaction costs as all these strategies

Table 5: RMSFE of forecasting models with alternative fair-value

	FV-VAR1	FV-VAR2
All	3.57**	3.48**
pre-1990	2.26***	2.03***
post-1990	3.94**	3.88**

The table reports the RMSFE of each forecasting method. *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. Data used to estimate the models and to perform out-of-sample forecast evaluation is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

can be implemented with low-cost and liquid ETF products, and all the strategies would be affected by the same costs.

Overall the economic gains tend to be consistent with the statistical gains analysed earlier. The FV-VAR2 model performs best among the alternatives analysed, with statistically significant differences with respect to the historical average method for any type of investor. Starting from the case of an investor with risk aversion 2 (first row), all methods generate utility gains with respect to the historical average method apart from the VAR1 method, with annaulised 0.11% and 0.63% gains for the CAPE and EY methods respectively, and 0.76% of the FSC2011, however only the FSC2011 results are statistically significant at 10% level. Moving on to the VAR methods, the VAR1 generates a utility loss, while the VAR2 method generates a utility gain of 0.92% (higher than the FSC2011 by roughly 20 bps). Both FV-VAR methods generate the highest utility gains among all the methods: FV-VAR1 generates slightly more than 2.4% CER gains every year, while FV-VAR2 generates 2.75% CER gains every year. Both results are also statistically significant at 5% confidence level.

Looking at the case of a risk-seeking investor that implements her strategy without short or leveraged positions (second row), results are consistent with the unconstrained strategy although with slightly lower utility gains. FV-VAR

Table 6: CER gains over historical average method

	CAPE	EY	FSC2011	VAR1	VAR2	FV-VAR1	FV-VAR2
$\gamma = 2$	0.11	0.63	0.76*	-0.27	0.92*	2.4**	2.75**
$\gamma = 2, \text{const.}$	-0.06	0.26	0.78*	-0.08	0.67*	1.9**	2.2***
$\gamma = 4$	-0.33	0.05	-0.21*	-0.93	-0.32*	0.62***	0.75***

The table reports the difference between certainty equivalent return (CER) of optimal portfolios built using each forecasting method indicated in columns and the CER of optimal portfolios built using the historical average forecasting method. Investor is assumed to optimise CRRA utility function with static portfolio weights and a 10-year investment horizon. Columns refer to various forecasting methods: *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields and macroeconomic volatility estimated with a steady-state Bayesian VAR. Rows refer to various investor's profiles: $\gamma=2$ is the case of an investor with risk aversion equal to 2, $\gamma=2, \text{const.}$ is the case of an investor with risk aversion equal to 2 and risky asset weight constrained between 0% and 100%, $\gamma=4$ is the case of an investor with risk aversion equal to 2. Data used to estimate the models and to perform out-of-sample forecast evaluation is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between CER of each method versus the CER of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

strategies perform better than other methods and are the only ones generating results that are statistically different at 5% confidence level with respect to the historical average method.

Finally, looking at the case of a more risk-averse investor (third row), utility gains for all methods decrease compared to previous types of investors. This is not a surprise since the goal of our methods is to improve the forecasting accuracy of equity returns and this is also in line with results presented in the literature (e.g. Faria and Verona 2016, among others). However, despite lower nominal levels, CER gains for FV-VAR continue to be the highest among all methods tested and are the only ones to be statistically different from the historical average method at 1% confidence level.

In conclusion, these results confirm that different types of long-term investors are better off using forecasting models such as FV-VAR1 and FV-VAR2 that combine signals for the three components of equity returns and that take into

account changing macroeconomic conditions in order to model future paths of equity valuations.

1.5 Conclusions

In this paper I explore the impact of using information from a fair-value model of equity valuations in order to forecast future 10-year US equity returns. Using the fair-value model to inform the steady-state prior for US earning yield modelled in a Bayesian VAR a la Villani (2009), I extend the Ferreira and Santa Clara (2011) method to forecast long-term equity returns and I show that this framework produces statistically and economically significant improvements with respect to competing methods such as simple historical average, popular predictive regressions, and the Ferreira and Santa Clara (2011). In particular, the proposed framework reduces the out-of-sample root mean square forecast error of 10-year US equity returns by annualised 1.2% with respect to the Ferreira and Santa Clara (2011) method and by annualised 2.5% on a model based on simple historical average. Moreover, for investors that are willing to take on investment risk, it generates out-of-sample certainty equivalent return gains for a long-term investor of annualised 2% with respect to the Ferreira and Santa Clara (2011) method and of annualised 2.75% on a model based on simple historical average.

In terms of future research, it would be interesting to investigate more the assumptions made for another important component of equity returns: earnings growth. While in this paper we followed the approach proposed by Ferreira and Santa Clara (2011) of forecasting earnings growth using their 20-year moving average, the volatility of earnings after the 2007-09 crisis generates large swings in equity return expectations and, as a consequence, portfolio weights. Also, earnings growth has historically been cyclical, therefore it would be interesting to test if alternative assumptions could lead to higher forecast accuracy of total equity returns and less volatility in portfolio weights for an investor that uses those forecasts to build an equity-bond portfolio.

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2 Paper 2: Modelling earnings growth to forecast long-term US equity returns

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2.1 Introduction and literature review

Earnings of companies are a fundamental driver of equity returns. In essence, they are the source of value for equity investments (Cornell 2010) as investors pay a price to acquire the right to participate in the distribution of those companies' earnings. At the same time, the price of companies' shares paid by investors depends on aggregate expectations of future earnings: earnings are therefore central to firm valuation and are the basis for the accounting-based cost of capital calculations. Therefore, producing accurate estimates of future earnings growth is crucial for investors.

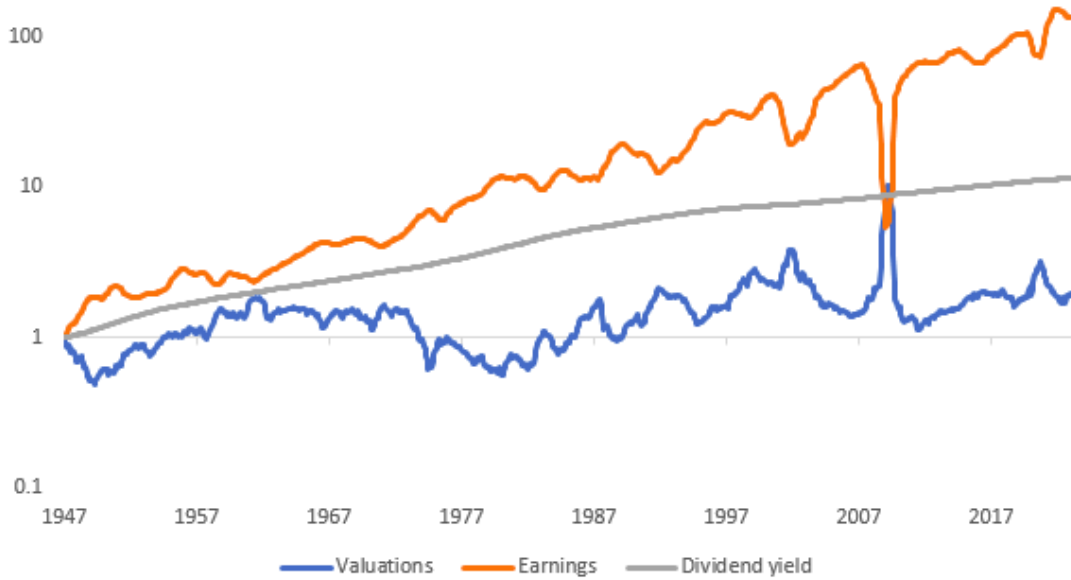
To better analyse the importance of companies' earnings growth for equity returns over the short, medium and long-term time horizon, it is useful to decompose equity returns into their structural components, as suggested by Ferreira and Santa Clara (2011) among others: an income return component, namely the dividend-price ratio (dy) in the case of equities, and a price return component that can be further decomposed into earnings growth (Δe) and growth of the price-to-earning ratio (Δpe).

$$(1 + r) = (1 + \Delta e) * (1 + \Delta pe) * (1 + dy) \quad (1)$$

To start, it is interesting to see how each of these components affect long-term equity performance. Using data from March 1947 till March 2023, Figure 1 reports the cumulative return of each of the S&P500 index return components over time. Values are reported in logarithmic scale of base 10 to facilitate the visualisation. While equity valuations tend to be mean-reverting and therefore their long-term

contribution fluctuates around the zero line, the majority of the historical total equity return is explained by dividend yield and, more significantly, by earnings growth that is by far the most important contributor.

Figure 1: Cumulative returns of the components of S&P500 total returns from 1947 to 2023



This figure reports the cumulative wealth generated by each of the components of total equity returns identified in Equation 1. The orange line is the cumulative return of earnings growth (Δe), the gray line is the cumulative return of the dividend-price ratio (dy), the blue line is the cumulative return of growth of the price-to-earning ratio (Δpe). Data used is from March 1947 to March 2023, and it is downloaded from Shiller's database and the Global Financial Data database. The vertical axis reports values in logarithmic scale with base 10.

In numbers, from March 1947 till March 2023, the average annaulised total return of the S&P500 index is equal to 11.2%, with 62% of the total return explained by earnings growth, 30% explained by the dividend component, and the remaining 8% by growth in valuations.

While characteristics of returns over the long-term are important, in reality investors build financial plans and take asset allocation decisions based on medium-term horizons, usually 10 years. Focusing on this time horizon, Figure 2 reports the rolling 10-year annualised total returns of the S&P500 index along with 10-year annualised income returns, driven by the dividend yield, and price returns, driven by both changes in valuations and earnings growth. As we can see, the variation of

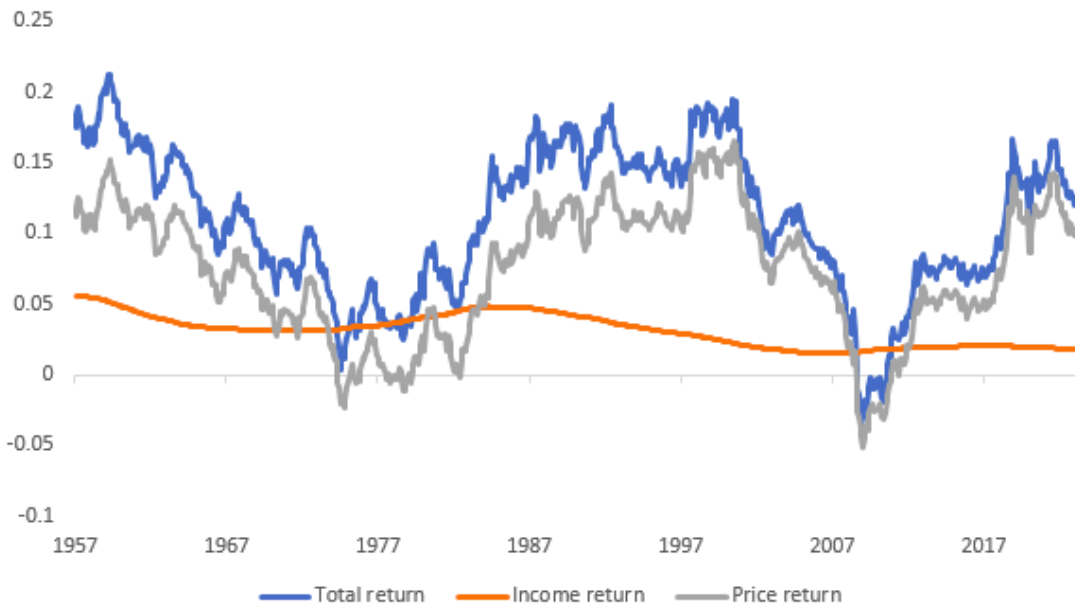
Table 1: Contribution of the components of S&P500 total returns (annualised)

	Tot. Ret.	Δpe	Δe	dy
Return	11.2%	0.8%	6.7%	3.3%
% of Tot. Ret.		7.8%	61.8%	30.3%

The table reports the annualised contribution of changes in valuations, earnings growth and dividend yield to S&P500 total returns. The first row reports the annualised return of variables reported in column headers. The second row reports the ratio (in %) of each return component to the total return of the index. Data used is from March 1947 to March 2023, and it is downloaded from Shiller's database.

medium-term US equities returns is primarily explained by the variation in prices.

Figure 2: 10-year total return components: income return and price return

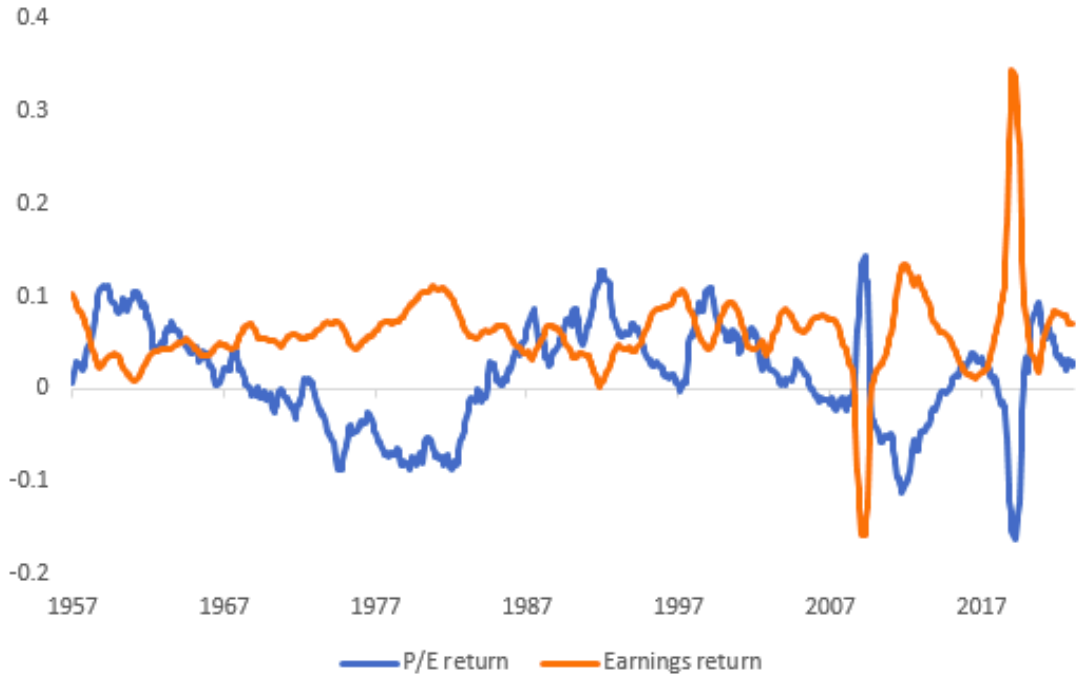


The blue line is the rolling annualised 10-year total return of the S&P500 index, the gray line is the rolling annualised 10-year price return of the S&P500 index, the orange line is the rolling annualised 10-year income return of the S&P500 index. Data used is from March 1947 to March 2023, and it is downloaded from Shiller's database and the Global Financial Data database.

Next, to better understand the nature of the variation in price returns, Figure 3 reports the rolling 10-year annualised return of changes in valuations and earnings growth of the S&P500 index. As we can see, both components significantly affect the time-variation of equity returns and both tend to show a cyclical pattern. In particular, while earnings growth had a positive long-term average in line with results reported in Table 1, periods of higher earnings growth have been followed by periods of lower earnings growth. On the other hand, changes in valuations

have also been mean-reverting historically, contributing to positively or negatively affecting equity returns over time.

Figure 3: 10-year price return components: valuations and earnings growth



The blue line is the rolling annualised 10-year growth return of the price-to-earning ratio of the S&P500 index, the orange line is the rolling annualised 10-year growth return of the earnings-per-share of the S&P500 index. Data used is from March 1947 to March 2023, and it is downloaded from Shiller's database and the Global Financial Data database.

Given the results presented so far, it is easy to understand why there exists an extensive literature of academic and investment practitioners' studies aimed at forecasting both changes in equity valuations and earnings growth: more accurate forecasts of future valuations and earnings growth potentially translate into more accurate forecasts of total equity returns. In particular, changes in valuations and how they contribute to forecast long-term equity returns have been analysed in several studies such as Fama and French (1988, 2002), Bogle (1991), Cochrane (2008). More recently, Cilla (2023) extended the Ferreira and Santa Clara (2011) framework to allow for long-term reversal of valuations to a steady-state based on macroeconomic conditions.

At the same time, a growing body of academic and practitioners' research

has been dedicated to analysing and forecasting earnings growth. Today, one of the most popular sources for obtaining earnings forecasts are financial analysts. These forecasts are aggregated by data providers, such as the Institutional Brokers Estimate System (I/B/E/S), and subsequently made available to academics and investors. Whilst popular, researchers over time have found a significant optimism bias in those estimate, see for example Francis and Philbrick (1993), McNichols and OBrian (1997), Easton and Sommers (2007).

Alternatives to analysts earnings forecasts are econometric models, normally built on regression-based frameworks. They can be time-series models that therefore rely on past realizations of earnings or on a combination of past earnings and other financial factors. Or they can be corss-sectional models, therefore aimed at explaining variation of earnings at a given point in time, using a set of explanatory variables. Starting from cross-sectional models, Hou et al. (2012) developed a model based on assets, earnings, and dividends, which however underperformed analysts forecasts in terms of forecast accuracy. Gerakos and Gramacy (2013) find that a simple random-walk model, in which the most recent available value is used to forecast future values, performs as well as other, more sophisticated, earnings forecast models. Also, Li and Mohanram (2014) develop a forecasting framework based on the so called earnings persistence, that is the extent to which current earnings persist or recur in the future, and the so called residual income, that is the amount of leftover operating profit after paying all costs of capital used to generate the revenues. They show that these models are superior to the Hou et al. (2012) and random-walk models in terms of forecast accuracy.

At the same time, a large body of research has examined the forecast accuracy of regression-based time-series models and generally concluded that they cannot match the forecast performance of professional analysts, especially at short forecast horizons of one quarter or less. The empirical evidence for the superiority of analysts forecasts suggests that analysts perform better because they have access to a broader spectrum of information that is frequently observed (i.e., they have

both an informational and a timing advantage), as suggested by Ball and Ghysels (2017). Examples of studies in this area include Brown and Rozeff (1978), Collins and Hopwood (1980), Fried and Givoly (1982), Brown et al. (1987), O'Brien (1988), and Bradshaw et al. (2012). This is broadly consistent with Campbell and Shiller (1998), Fama and French (2002), and Cochrane (2008) that find that earnings growth is nearly unforecastable. More recently, Ball and Ghysels (2017) developed a model based on mixed data sampling regression methods (MIDAS), which combines various high-frequency time-series data to forecast earnings. Their model outperforms raw analysts forecasts in some cases and they also find that it can be combined with analysts forecasts to improve forecast accuracy. Their model is, however, limited to short-term forecast horizons (next quarter in particular), and therefore it is not suited to eventually forecast long-term equity returns. Focusing on longer horizons, Bradshaw et al. (2012) find that earnings per share forecasts from a simple random walk model are more accurate than analyst forecasts over longer horizons. At the same time, Binsbergen and Koijen (2010) show that earnings growth has a low-frequency predictable component possibly due to a change over time in macroeconomic variables, in particular inflation since earnings growth is a nominal variable. Based on these findings, Ferreira and Santa Clara (2011) propose a 20-year moving average of earnings growth to forecast earnings growth and consequently equity returns up to 5 years.

In line with the intuition shared by Binsbergen and Koijen (2010), another stream of literature has analysed the link between the macroeconomy and companies' earnings, and consequently the link between economic growth and equity returns. As stated by Cornell (2010), the performance of equity investments in the aggregate is inextricably linked to economic growth. At the same time, earnings are themselves driven by aggregate economic activity. In particular, unless corporate profits rise as a percentage of GDP, something that cannot continue indefinitely, earnings growth is constrained by GDP growth. This means that the same factors that determine the rate of economic growth will also place bounds on earnings growth and, thereby, the performance of equity investments. Cornell (2010) concluded that the long-run performance of equity investments is

linked to the earnings growth, which depends on GDP growth over time. Moreover, he finds that, over the long run, aggregate earnings are a stationary fraction of GDP and that, while the earnings-GDP ratio is stationary, it is far from constant. In fact, when earnings are low relative to GDP they grow more quickly with the reverse occurring when earnings are relatively high. It should be noted though that in an efficient market, the mean reversion in earnings growth should have no impact on stock returns because it would be already included into current prices. However, Campbell and Shiller (1998) provide evidence that long-run average earnings can contribute to predict future stock returns. Specifically, when the ratio of price to the average earnings over the previous ten years is high, future stock returns tend to be low and the reverse is true when the ratio is low. This suggests that the market does not fully account for the mean reverting nature of long-run earnings growth.

At the same time, understanding the link between GDP growth and earnings growth in a given country is not an easy task. The fact that the ratio of aggregate earnings to GDP is stationary implies that investors can expect aggregate real earnings growth to be in line with real GDP growth in the long run. Unfortunately, the same is not true of the earnings to which current investors have a claim. There are two reasons for this discrepancy. First, an investors pro rata portion of a companys earnings will be affected by company share issuances and repurchases. If this dilution (or accretion) is on-going, growth in aggregate earnings and earnings per share will diverge. Second, current investors do not participate in the earnings of new businesses unless they dilute their current holdings to purchase shares in start-ups. This drives a wedge between the growth in aggregate earnings and the growth in the earnings to which current investors have a claim. These issues have been studied by Bernstein and Arnott (2003) who suggested a procedure for estimating the combined impact of both effects on the rate of growth of earnings-per-share. They note that total dilution on a market-wide basis can be measured by the ratio of the increase in market capitalization to the value-weighted increase in stock price. More precisely, each period net dilution is equal to $(1+c)/(1+p)-1$, where c is the percentage capitalization increase and p is the percentage increase

in the value weighted price index. They find that, from 1926 to 2008, earnings-per-share grew on average 2% a year slower than the economy.

Another cause of mismatch between earnings of listed companies in a given country and economic growth of the same country is the effect of globalisation and the internationalisation of enterprises.⁸. In this context it is important to note that the GDP measure follows the national accounts principle, that is it captures only the operating profit of firms carrying out their activities within the respective economic territory and thus does not include the profit of multinationals overseas subsidiaries. At the same time, data on the geographical breakdown of the S&P500 companies earnings suggest that around 30% of total earnings of companies included in the index come from international (non-US) markets.⁹ To sum up, while in theory companies' earnings and GDP should be connected, it is important to carefully take into account the aspects mentioned above in order to make a meaningful comparison.

Starting from these findings, the goal of this paper is to contribute to the literature of forecasting long-term earnings growth by using time-series models and by conditioning the future evolution of earnings growth to macroeconomic conditions. In particular, I adapt a steady-state Bayesian VAR model a la Villani (2009) in order to forecast earnings growth over the next ten years, using a measure of long-term trends in GDP growth as a prior for the steady-state of the endogenous variable included in the system. In my case, the Bayesian VAR reduces to a Bayesian AR model as, for simplicity, I only use one endogenous variable: earnings growth. Moreover, the GDP growth measure used as steady-state prior is appropriately adjusted to take into account the effects due to dilution and internationalisation of companies, documented in the literature. Furthermore, this paper contributes to the literature of forecasting long-term US equity returns, extending Ferreira and Santa Clara (2011) and Cilla (2023) by replacing the 20-year moving average assumption used to forecast future earnings, with the earning

⁸Among other sources, this is analysed in the ECB Monthly Bulletin of September 2007 available at <https://www.ecb.europa.eu/pub/pdf/mobu/mb200709en.pdf>

⁹The regional breakdowns of companies earnings are computed using FactSet data and are as of December 2022.

growth forecasts generated with the Bayesian VAR with steady-state priors. The proposed framework improves out-of-sample forecast accuracy of earnings growth by 1% annualised compared to the method proposed by Ferreira and Santa Clara (2011) based on the 20-year rolling average of earnings growth, and by 1.6% annualised with respect to an autoregressive method whose forecasts converge to the long-term historical average. In terms of equity return forecasts, the proposed method helps to reduce the out-of-sample root mean square forecast error (RMSFE) of US equity returns over an horizon of 10 years by further 35 basis points annualised with respect to the FV-VAR models presented in Cilla (2023). It also helps to generate utility gains for investors that overall are in line with the FV-VAR frameworks: slightly higher than the FV-VAR1 and slightly lower than the FV-VAR2. Finally, the results show a significant impact of earnings growth volatility in the periods post-Global Financial Crisis.

Next sections are organized as follows. In section 2, I provide the details of the methodology used to define a steady-state prior for long-term earnings growth as a function of GDP growth. In section 3, I explain how I use the Villani (2009) framework to forecast earnings growth and consequently equity returns. In section 4, I present the empirical results related to the accuracy of return forecasts and the economic gains for long-term investors that would use the proposed method to allocate their capital between US equities and US bonds. Section 5 contains concluding remarks.

2.2 Methodology - Measuring earnings growth and the links to the economy

The first aspect to note in order to analyse companies' earnings and their link to the economy is that there are various measures of earnings that have been studied by the literature and are constantly monitored by investors: on one hand, reported earnings and operating earnings, that are commonly computed for equity indices such as the S&P500; on the other hand, national income and product accounts (NIPA) earnings produced and published by the US Bureau of

Economic Analysis (BEA). As noted in Petrick (2001), the differences between the NIPA profit measures and the S&P 500 earnings measures reflect differences in purpose, definitions, and methodologies. In fact, the S&P earnings measures are broad market measures that serve as benchmarks for assessing the performance of individual companies or groups of companies, while the NIPA profits measures serve as time series that provide consistent coverage of all the Nations corporations. In particular, the S&P 500 measures of profits - reported earnings, operating earnings, and their per-share versions - reflect the aggregate earnings of the 500 corporations that compose the S&P 500 stock index, and they are measured on a financial accounting basis: reported earnings are based on the after-tax earnings that are publicly reported by corporations; operating earnings are reported earnings that exclude the impact of cumulative accounting changes, discontinued operations, extraordinary items, and special items. It is important to note that operating income is not defined by the Financial Accounting Standards Board and, as reported in Siegel (2016), this permits individual companies to interpret what is and what is not 'extraordinary' or 'special'. The result is a varied interpretation (by companies) of items and charges, where the same specific type of charge may be included in operating earnings for one company and omitted from another. On the other hand, corporate profits in national income is the income earned from current production by corporations. Since national income is defined as the income of U.S. residents, its profits component includes income earned abroad by U.S. corporations less income earned in the United States by foreign corporations. Income consists of receipts that arise from current production less associated expenses. Receipts exclude income in the form of dividends and capital gains, and expenses exclude bad debts, depletion, and capital losses.

While the long-term trends of the NIPA profits measures and the S&P profits measures are roughly similar, the year-to-year changes sometimes differ substantially. This is particularly true during recessions when a key source of high volatility in S&P 500 quarterly reported earnings is asset writedowns. As documented in Hodge (2011), in the most extreme case, in the fourth quarter of 2008, S&P 500 companies reported a loss of \$202.1 billion (not annualized) after

a third-quarter profit of \$65.2 billion. This was associated with the writedowns of prominent financial companies: AIG lost \$61.7 billion, and large writedowns were also recorded by other large financial companies in the S&P 500, including Citigroup, Bank of America, and Merrill Lynch. In the fourth quarter of 2002, S&P 500 reported earnings fell 64.1 percent or \$51.0 billion. This roughly accords with the \$45.0 billion loss reported that quarter by Time-Warner, at that time the largest quarterly corporate loss in history. At the same time, S&P 500 operating earnings in the fourth quarter of 2008 turned down, to a loss of \$0.8 billion from a gain of \$87.8 billion in the third quarter. Although writedowns are excluded from the S&P 500 operating earnings measures, trading gains and losses are considered part of S&P 500 operating profits and losses, and a portion of these are likely capital losses on held positions rather than spread or market-making profits. NIPA profits indicators, such as the Quarterly Financial Report and Quarterly Banking Profile, may exclude these losses more effectively and NIPA profits may give more weight to industries that did not experience large capital losses. For these reasons, Siegel (2016) proposed to use either operating or NIPA series of companies' earnings to assess the performance of the US equity market, and in particular he proposed a new version of the CAPE ratio built using NIPA and operating earnings instead of reported earnings as in Campbell and Shiller (1988). Given the scope of this paper is to ultimately forecast S&P500 equity returns, I follow Campbell and Shiller (1988), Ferreira and Santa Clara (2011), and Shiller (2021) by relying on the S&P500 measures from now on, using the reported earnings as base case and operating earnings to perform robustness checks. For all analyses, reported earnings data is downloaded from Prof. Shiller database. Operating earnings are published by Standard and Poor from December 1988 therefore, to test the impact of using operating earnings instead of reported earnings, I follow Siegel (2016) and build a spliced earnings growth series using operating earnings starting in December 1988 and reported earnings before December 1988.

Once analysed and chosen the appropriate earnings measures, it is now possible to analyse whether companies' earnings have historically co-moved with the overall economy. To start, it is important to recognise that characteristics of the aggregate

corporate earnings might not be the same of the earnings to which current investors have a claim. In fact, while total earnings are an important measure as they track the overall performance of corporates, investors can only buy a limited number of shares of those corporates. For this reason, what matters for investors are earnings-per-share, for which both the total earnings at the numerator and the number of shares at the denominator play an important role. In particular, as reported in Bernstein et al. (2003), while there are structural reasons why earnings and the overall economy should grow at the same rate over the long-term, earnings-per-share can keep up with GDP only if no new shares are created. Entrepreneurial capitalism creates a dilution effect that is the result of the creation of shares as existing and new companies capitalize their businesses with equity. At the same time, companies can also repurchase outstanding shares with the aim to reduce the number of shares in the market and increase the value of remaining shares, the so called share buyback, that therefore has the opposite effect of stock new issuances. Stock buybacks are attractive to companies and beneficial to investors as they are a tax-advantaged ways of providing a return on shareholder capital and preferable to dividends which are taxed at a higher rate.

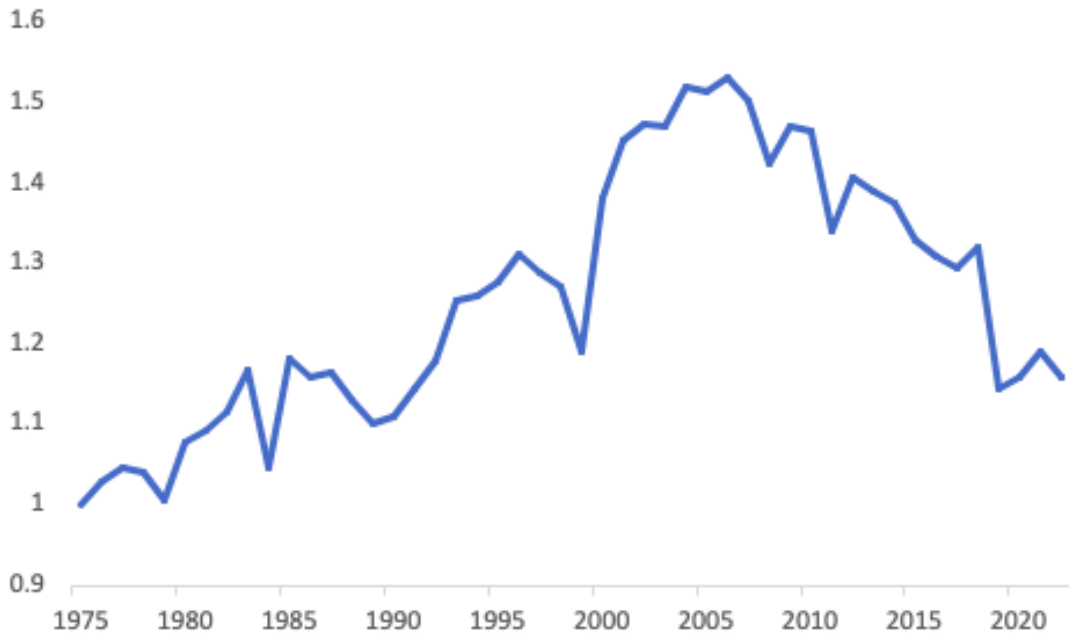
To understand the combined effect of both share dilution and buybacks on earnings-per-share, Bernstein et al. (2003) developed a measure aimed at analysing the volume of *net* buybacks, that is stock buybacks less new share issuance. To do so, they develop a measure equal to the the ratio of the proportionate increase in market capitalization to the proportionate increase in stock price. Formally, they define the net dilution measure as:

$$Net\ dilution = \frac{(1 + c)}{(1 + r)} + 1 \quad (2)$$

where c is the percentage capitalization increase, and k is the percentage increase in the value weighted price index. This relationship has the advantage of factoring out valuation changes, which are embedded in both the numerator and denominator, and neutralizing the impact of stock splits. However, it is important to stress that this measure holds only for broad market indexes, such as the CRSP 1-10 or the Wilshire 5000, because less inclusive indexes can vary

the ratio simply by adding or dropping securities. To give a better idea of how this dilution has proceeded historically, Figure 4 plots the dilution index, defined as the ratio of capitalization growth to price index growth for the total US equity market here represented with the Wilshire 5000 index. For this purpose, the market capitalization of US listed domestic companies is downloaded from the World Federation of Exchanges database, while the price index of the Wilshire 5000 index is downloaded from FRED Federal Reserve Bank of St. Louis database.

Figure 4: Dilution rate



The blue line is the time series of the dilution measure defined as: $(1 + c)/(1 + p) - 1$, where c is the percentage capitalization increase of total US listed domestic companies downloaded from the World Federation of Exchanges database, and p is the percentage increase in the value weighted price of the Wilshire 5000 index downloaded from FRED Federal Reserve Bank of St. Louis database.

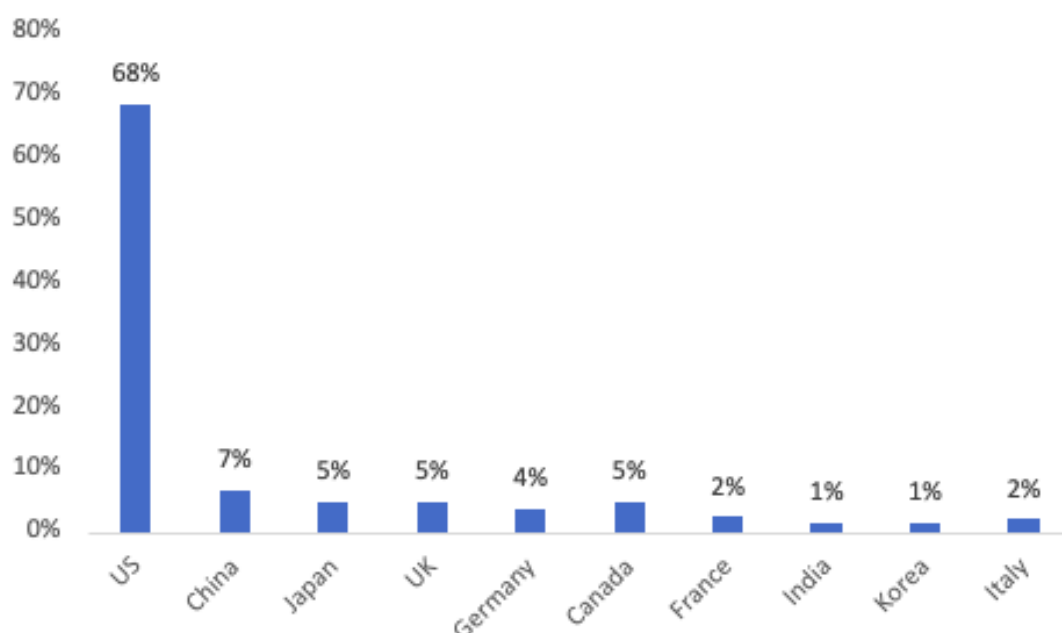
The fact that this line rises nearly monotonically till 2007 shows that issuance of new shares almost always exceeded stock buybacks in that period, as shown in Bernstein et al. (2003). This is the case for most of the years before 2007 with some notable exceptions occurred in the late 1980s and late 1990s, when buybacks modestly outpaced new share issuance. At the same time, the overall trend started to reverse after the beginning of 2007 when the net dilution measure

starts to decrease at a steady rate till today. According to analysts¹⁰, this has likely been the result of higher amount of cash availability by companies thanks to low interest rates and tax incentives to repatriate capital from abroad, that is then been distributed to shareholders via buybacks. Whatever the exact cause of this trend, this proves that a more dynamic approach than the constant 2% rate suggested by Bernstein et al. (2003) is needed in order to take into account changes in long-term trends.

A third aspect that is important to consider when comparing corporate earnings to economic growth is that, over the years, US corporations have increased their international presence due to globalisation that created opportunities to sell goods and services internationally. In particular, today around 30% of the S&P500 companies' earnings is produced in countries other than the US. Figure 5 reports the average percentage of profits generated in the US versus other countries, from 2003 till 2023. Note that countries have been selected so that 90% of total geographical exposure is reached and then rescaled to sum 100%.

¹⁰e.g. <https://www.ft.com/content/aff56c18-ecc8-33c9-bdd8-382ca72bb0b7>; <https://www.ft.com/content/f6f59092-7ff5-4095-ba25-084bcc298c24>; <https://www.ft.com/content/1c59a9b0-5c14-4981-84b8-d0007179ff3c>

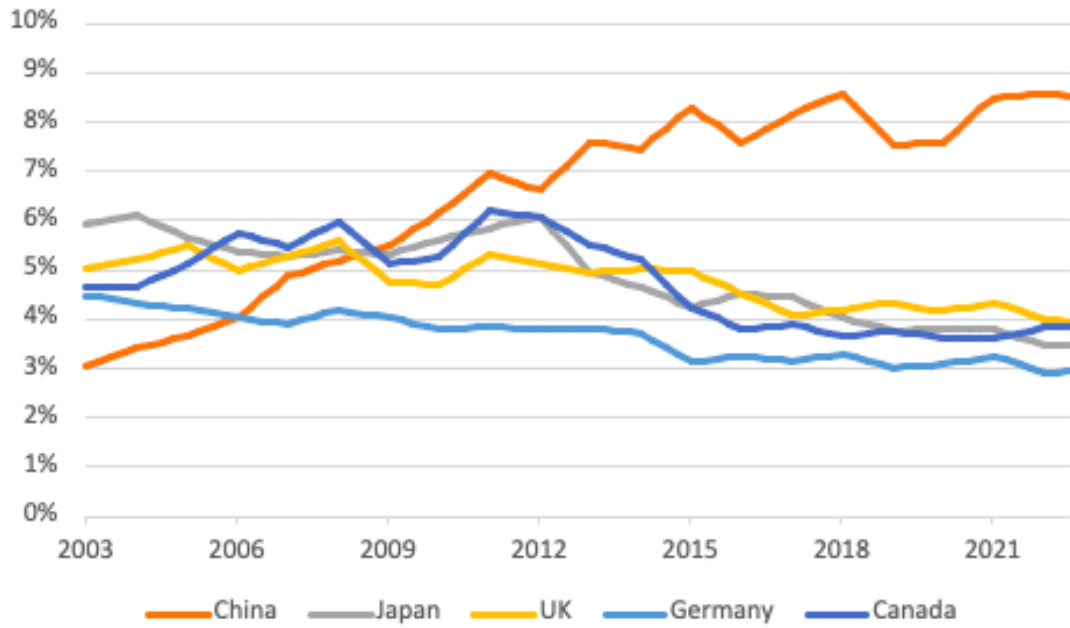
Figure 5: Average earnings geographical exposure, from 2003 to 2023



Each bar represents the average percentage of profit of S&P500 index companies generated in countries reported on the x-axis, from 2003 to 2023. Data is downloaded from FactSet.

While average exposures already give interesting insights, it is also important to analyse how exposure to various countries has changed over time. Figure 6 reports the geographical exposure of S&P500 companies' earnings in countries other than the US, and in particular for countries that have had at least 3% weight since 2003. It is interesting to observe the increasing role that the Chinese economy has played for US companies compared to other countries over time: China weight grew from 3% in 2003 to almost 9% in 2023, while other economies such as Japan or the UK have seen a decrease in importance. This is justified by the increasing role that China has played in the global economy, especially after China entered the World Trade Organisation in 2001. It is therefore likely that the presence of US corporates in emerging markets like China, characterised by growth rates higher than more developed economies, have contributed to boost US companies' profits especially over the more recent years.

Figure 6: Earnings geographical exposure outside the US



Each lines represents the time series of the percentage of profit of S&P500 index companies generated in various countries, from 2003 to 2023. Data is downloaded from FactSet.

At the same time, it is important to note that the GDP measure follows the national accounts principle, that is it captures only the earnings of firms carrying out their activities within the respective economic territory and thus does not include the profit of multinationals overseas subsidiaries. As economic globalization continues and more firms operate in several locations throughout the world, it is important to take this data into account when comparing corporate earnings with GDP measures.

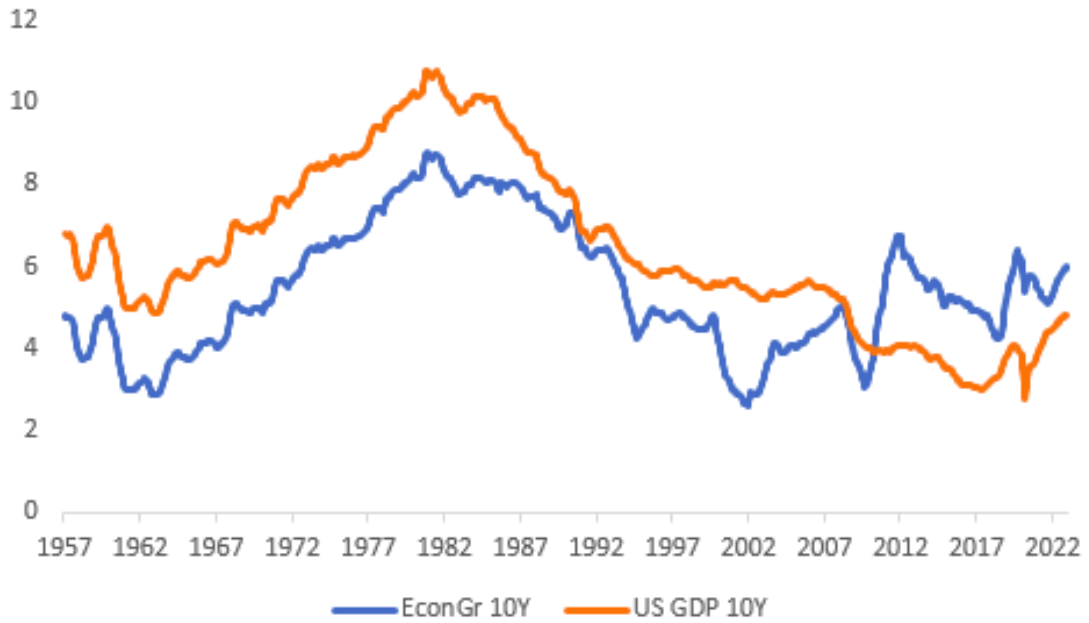
Therefore, taking into account the effects due to share dilution, share buyback and international exposure of US companies, I develop a measure of economic activity that is consistent with earnings-per-share dynamics of S&P500 companies. In particular, I start with the US nominal GDP measure and I correct this measure for the net dilution effect using the measure defined in equation 2. Moreover, while before 2003 only US GDP is used, after 2003 when the data on geographical exposure of corporate earnings starts, I compute a weighted GDP measure using the USD-GDP series of all relevant counties presented in Figure 5. GDP data is downloaded from the FRED Federal Reserve Bank of St. Louis database and the

Global Financial Data database. I define the economic measure as:

$$Economic\ Growth_t = \begin{cases} \Delta GDP_{t,US} + \Delta NetDilution_t & \text{up to 2003} \\ \sum_{i=1}^n \Delta GDP_{t,i} * w_{t,i} + \Delta NetDilution_t & \text{after 2003} \end{cases} \quad (3)$$

where $\Delta GDP_{t,i}$ is the growth rate of the nominal GDP of the country i at time t , $\Delta NetDilution_t$ is the growth rate of the net dilution measure defined in equation 2 at time t , $w_{t,i}$ is the weight of the US companies revenue exposure to country i at time t . The weight of geographical revenue exposure of S&P500 companies is downloaded from FactSet for the following countries: US, China, Japan, UK, Germany, Canada, France, Italy, Republic of Korea, India, that overall explained 90% of the total earnings' geographical exposure of S&P500 companies since 2003. This measure will serve as the economic-based equilibrium for earnings-per-share (EPS) growth. Figure 7 reports the 10-year annualised growth rate of the economic measure versus the pure US GDP measure. While before early 2000s the EPS economic measure lags the US GDP growth rate by 2% on average in line with Bernstein et al. (2003) findings, the overall trend starts to revert after 2003 when growth in emerging markets like China contributes to boost US companies' earnings and, in particular, the 10-year trend of the EPS economic measure surpass the US GDP growth rate after 2009 also thanks to a negative net dilution effect, therefore with a positive effect on earnings-per-share.

Figure 7: EPS-consistent economic growth measure versus US GDP growth, rolling 10-year annualised

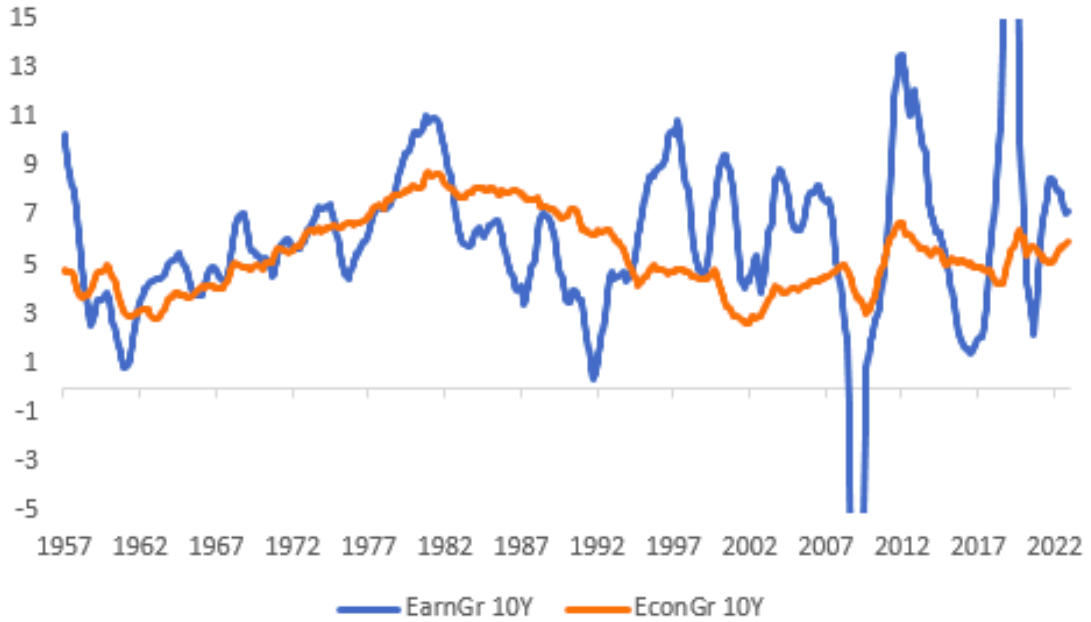


The blue line is the rolling 10-year annualised average growth rate of the economic measure defined in Equation 3, the orange line is the rolling 10-year annualised average growth rate of US GDP. GDP data is from FRED Federal Reserve Bank of St. Louis database.

It is also important to see how the EPS-consistent economic growth measure compares to the growth of earning-per-share itself. Figure 8 reports 10-year annualised rolling averages of both variables from 1957 to 2023, using reported earnings to measure the earning-per-share. To ensure readability, the vertical axis of the chart has been truncated between -5% and +15%, avoiding to show values of -15% reached during 2008-2009, and +30% during 2018-2019 as a result of the volatility of earnings reached during the Global Financial Crisis. As we can see, the average trend of reported earnings fluctuates around the economic measure defined in Equation 3, signalling that this measure potentially represents a valid economic-based equilibrium value for earnings-per-share. It is also interesting to note that earnings growth is sometimes above or below the equilibrium: in particular periods in which the trend EPS growth is higher than the economic equilibrium are followed by more muted EPS growth rates, and viceversa. Combining the information shown in Figure 8 with Figure 7, It is also clear that using pure US GDP growth would have represented a biased equilibrium measure for EPS growth, as suggested by the literature, being too high for the years before early

2000s and too low for the trends experienced by earnings afterwards.

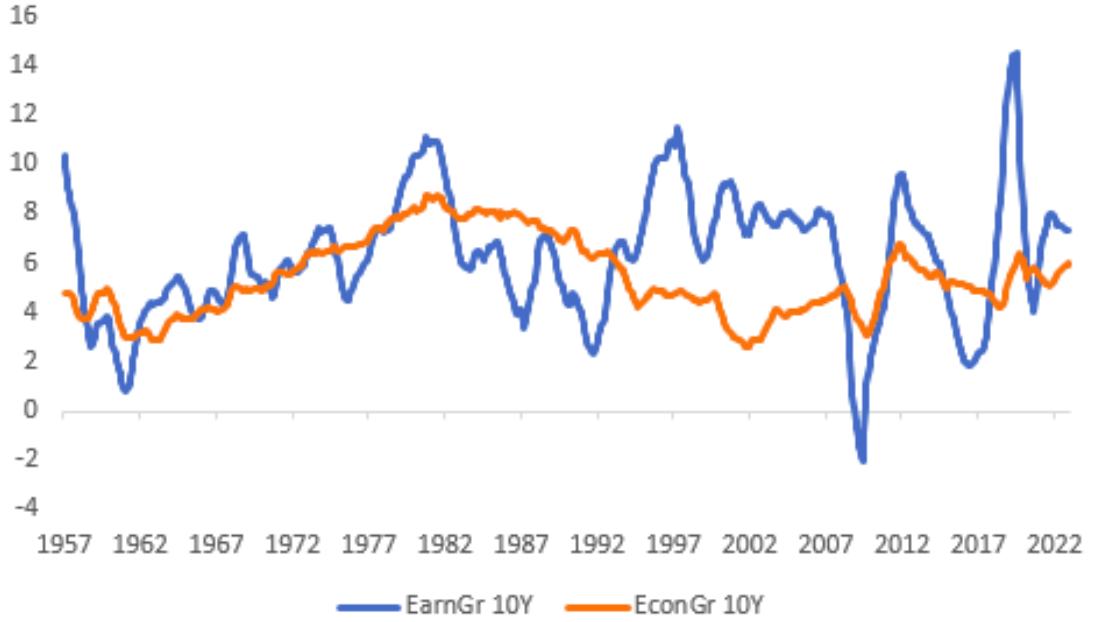
Figure 8: Reported earnings growth vs economic growth, rolling 10-year annualised



The blue line is the rolling 10-year annualised average growth rate of reported earnings-per-share of S&P500 companies, the orange line the rolling 10-year annualised average growth rate of the economic measure defined in Equation 3. Earnings-per-share data is from Shiller's database.

As a robustness check, Figure 9 replicates the results shown in Figure 8 using operating earnings instead of reported earnings. As we can see, results are consistent in the two cases, however with much lower volatility of EPS after early 2000s, in particular during recession periods such as the Global Financial Crisis.

Figure 9: Operating earnings growth vs economic growth, rolling 10-year annualised



The blue line is the rolling 10-year annualised average growth rate of operating earnings-per-share of S&P500 companies, the orange line the rolling 10-year annualised average growth rate of the economic measure defined in Equation 3. Earnings-per-share data is from Shiller's database and Standard and Poor's database.

2.3 Methodology - Forecasting earnings growth and equity returns

In order to generate equity return forecasts I extend the Ferreira and Santa Clara (2011) 'sum of the parts' and the Cilla (2023) frameworks. In particular, their methods are based on the intuition that equity returns (r) can be decomposed into three components: an income return component, namely the dividend-price ratio (dy) in the case of equities, and a price return component that can be further decomposed into earnings growth (Δe) and growth of the price-to-earning ratio (Δpe), as shown in Equation 1.

Ferreira and Santa Clara (2011) prove that by separately forecasting each component of equity returns it is possible to generate forecasts that have a lower root mean square forecast error and generate higher economic gains compared to popular forecasting methods based on a simple historical average of returns or the

so called predictive regressions, for periods up to five years. At the same time, Cilla (2023) shows that Ferreira and Santa Clara (2011) results are also valid for a forecasting horizon of 10 years, and that total return forecasts of the US equity market can be further improved both on a statistical and economic basis by forecasting the mean-reversion of the valuations component of returns (Δpe) to a fair-value informed by the level of real interest rates and macroeconomic volatility. While both Ferreira and Santa Clara (2011) and Cilla (2023) forecast future earnings growth using a rolling 20-year average, the scope of this paper is to further extend Cilla (2023) framework by also forecasting the mean-reversion of the earning growth component of returns (Δe) towards an equilibrium based on economic growth. In particular, since the scope of the paper is to forecast long-term equity returns, I model the variability of the 10-year trend of earnings growth over time, using the measure described in Equation 3 as a fair value: whenever the trend of earnings growth is higher or lower than the trend of the economic measure then it is reasonable to expect a reversion of earnings growth towards it.

2.3.1 Forecasting earnings growth

In line with Cilla (2023), in order to model the reversion of valuations to the proposed economic equilibrium defined in Equation 3, I use a steady-state Vector Autoregression (VAR) framework as in Villani (2009). Autoregressive models have long been used in the financial literature to forecast equity valuations and returns, e.g. Campbell and Viceira (2002) among others, producing forecasts for the endogenous variables that by definition converge to their long-term historical sample average. However, as Villani shows, departing from the mean-reverting assumption of the VAR forecasts and using prior information on the long-term convergence level of the endogenous variables included in the VAR can potentially lead to substantial gains in forecast accuracy.

Following Villani (2009), the steady-state Bayesian VAR system is defined as

follows:

$$\begin{aligned}\Pi(L)(x_t - \Psi) &= \epsilon_t \\ \epsilon_t &\sim N(0, \Sigma)\end{aligned}\tag{4}$$

where x_t is a vector of endogenous variables, Π is a lag polynomial defined as $\Pi(L) = I - \Pi_1 L - \Pi_2 L^2 - \dots - \Pi_p L^p$, p is the number of lags of the VAR. Following Villani (2009), the *random* variables Ψ represent the unconditional mean of the process as $E(x_t) = \Psi$. In our case, for simplicity, I only include the 10-year trend of earnings growth as endogenous variable in the VAR system, that therefore reduces to a single variable steady-state AR framework. As we can see, the long-term steady state level is treated as a model parameter that needs to be estimated. Given the Bayesian framework, the estimation starts by defining priors for each parameter, for which I follow Villani (2009).

In particular, the prior on Σ is taken to be:

$$p(\Sigma) \propto |\Sigma|^{-(p+1)/2}$$

The prior for $vec(\Pi)$ is a general multivariate normal distribution:

$$vec(\Pi) \sim N(\Theta_\Pi, \Omega_\Pi)$$

Then, the prior for Ψ is

$$vec(\Psi) \sim N(\Theta_\Psi, \Omega_\Psi)$$

In order to define the parameters of the prior distribution I follow Villani's approach. In particular, the prior proposed by Litterman (1986) is used on the coefficients in Π , with the default values on the hyperparameters in the priors advocated by Doan (1992), therefore the overall tightness is set to 0.2. Moreover, the prior mean on the first own lag in Π is set to 0.5. This is in line with the view that earnings growth should revert to the steady-state however with a convergence speed that is moderate: 0.5 is in fact half way between a prior close to 0, that

would indicate a very fast convergence, and a prior close to 1, that would indicate a very slow convergence. In my case, only the prior on the first own lag is needed as p is set equal to 1: this is justified by the fact that the first own lag already explains most of the variation of the trend of earnings growth over time.

Next, in order to estimate the posteriors, it is important to notice that the posterior distribution of the VAR system defined above is intractable. However, Villani (2009) shows that the posterior distribution of each set of model parameters conditional on the other parameters is tractable, and that a Gibbs sampler can be used to draw from the joint posterior. In particular, the conditional posterior of Σ is an inverted Wishart distribution:

$$\Sigma|\Pi, \Psi, D \sim IW(E'E, T)$$

where $E = \Pi(L)(x_t - \Psi d_t)$.

The posterior for $vec(\Pi)$ is:

$$vec(\Pi)|\Sigma, \Psi, D \sim N(\bar{\Theta}_\Pi, \bar{\Omega}_\Pi)$$

where $\bar{\Omega}_\Pi^{-1} = \Sigma^{-1} \otimes X'_\Psi X_\Psi + \Omega_\Pi^{-1}$, $\bar{\Theta}_\Pi = \bar{\Omega}_\Pi[vec(X'_\Psi Y_\Psi \Sigma^{-1}) + \Omega_\Pi^{-1} \Theta_\Pi]$, $Y_\Psi = [x_t - \Psi d_t]$, $X_\Psi = [x_{t-1} - \Psi d_{t-1}, \dots, x_{t-k} - \Psi d_{t-k}]$.

Then, the posterior for Ψ is

$$vec(\Psi)|\Pi, \Sigma, D \sim N(\bar{\Theta}_\Psi, \bar{\Omega}_\Psi)$$

where $\bar{\Omega}_\Psi^{-1} = U'(D'D \otimes \Sigma^{-1})U + \Omega_\Psi^{-1}$, $\bar{\Theta}_\Psi = \bar{\Omega}_\Psi[U'vec(\Sigma^{-1}Y'D) + \Omega_\Psi^{-1}\Theta_\Psi]$, $Y = [\Pi(L)x_t]$, $U' = (I_p q, I_q \otimes \Pi'_1, \dots, I_q \otimes \Pi'_k)$, $D = [d_t, -d_{t-1}, \dots, -d_{t-k}]$.

The posterior distribution of the steady state parameter is estimated by combining information included in the data and the prior information. For the steady-state variable in our framework, the prior comes from the economic

equilibrium defined in Equation 3. As in any Bayesian framework, the posterior distribution of the parameter will converge to the prior distribution as the uncertainty of the prior reduces. On the contrary, a wide distribution of the prior will lead the estimate of the posterior to be close to the information contained in the historical data. In my case, a tight steady state prior will cause the long-term forecast of the trend in earnings growth to be close to the estimated equilibrium, while a wider prior will yield forecasts of earnings yields that will converge to the long-term historical average. It is important to stress that for the steady-state VAR, convergence to the target distribution is in general fast unless the prior on the steady state is very weakly informative, in which case the steady state VAR should not have been used in the first place. Given the goal of this paper is to assess whether information from the fair-value model can improve earnings growth forecasts, I assume a tight distribution around the steady-state prior. In specific, at each time t , I use the fair-value estimate at time t to set the mean of the prior distribution and, to make the prior informative, I define the variance of the distribution by assuming a 95% probability for the earnings yield to be in the range of $\pm 0.5\%$ around the fair value. As an example, if at a given time the economic-based equilibrium suggests a level of earning growth trend equal to 4%, the steady-state prior is assumed to be Normally distributed with a mean of 4% and variance such that there is a 95% probability of being between 3.5% and 4.5%, where the variance is simply the sample variance estimated using all available historical data.

2.3.2 Forecasting equity returns

In order to generate equity return forecasts consistent with earnings growth forecasted from the steady-state VAR, I extend the Ferreira and Santa Clara (2011) and Cilla (2023) frameworks. It is important to note that Δpe and dy are linked through the payout-ratio (d/e) defined as the ratio of dividends over earnings. Therefore, starting from the decomposition of equity returns into dividend-price ratio (dy), earnings growth (Δe) and growth of the price-to-earning ratio (Δpe),

we get:

$$\begin{aligned}
(1 + r) &= (1 + \Delta e) * (1 + \Delta pe) * (1 + dy) \\
&= (1 + \Delta e) * (1 + \Delta pe) * \left(1 + \frac{d}{e} \frac{1}{pe}\right)
\end{aligned} \tag{5}$$

From Equation 5 we can conclude that the forecast of US equity returns r can be generated by forecasting earnings growth Δe , changes in price-to-earning ratio Δpe and the payout-ratio d/e . By making different assumptions on how to forecast each component it is possible to test how various methods perform in terms of forecast accuracy of total equity returns.

- *FV-VAR1*. Following Cilla (2023), this method forecasts the dividend-price ratio using the most recent available value at each point in time, therefore assuming it follows a random walk; at the same time, for changes in price-to-earning ratio I use the forecasts produced with the steady-state VAR introduced in Cilla (2023); in order to forecast earnings growth, I use the 20-year moving average of reported earnings-per-share of S&P 500 companies in line with Ferreira and Santa Clara (2011).
- *FV-VAR2*. It follows the same assumptions of *FV-VAR1* apart from the fact that the dividend-price ratio is forecasted using the forecasts of the price-to-earning ratio $1/pe$ from the steady-state VAR in line with the decomposition reported in Equation 5; while for the pay-out ratio d/e I simply use the full sample historical average up to the time in which the forecast is generated.
- *FV-VAR3*. In this paper I introduce a new method called *FV-VAR3* that is built using the same assumptions of *FV-VAR2* apart from the one related to earnings growth, that in this case is forecasted using the steady-state VAR defined in section 3.1.

Along with the proposed FV-VAR frameworks, I also forecast equity returns using alternative models popular in the literature and among practitioners.

- *Hist. Avg.* The first alternative model is the historical average of realised returns computed as the geometric annualised average of realised return up

to the time the forecast is generated:

$$r_{t+120} = \prod_{i=1}^t (1 + r_i)^{12/t} - 1$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, r_i are the monthly total returns of the S&P500 index at time i .

- *CAPE*. The second alternative model is based on the predictive regression defined as follows:

$$r_{t+120} = \alpha + \beta * CAPE_t + \epsilon$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, $CAPE_t$ is the Shiller's CAPE ratio at time t . The parameters α , β and ϵ are estimated using an expanding window OLS regression, with the time-series of rolling 10-year annualised returns as dependent variable and the time series of the Shiller's CAPE ratio lagged by 10 years as independent variable.

- *EY*. The third alternative model is similar to the CAPE method but replacing the CAPE ratio with the earnings yield ($1/CAPE$), computed as follows:

$$r_{t+120} = \alpha + \beta * EY_t + \epsilon$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, EY_t is the inverse of the Shiller's CAPE ratio at time t . The parameters α , β and ϵ are estimated using an expanding window OLS regression, with the time-series of rolling 10-year annualised returns as dependent variable and the time series of the inverse of Shiller's CAPE ratio lagged by 10 years as independent variable.

- *FSC2011*. The fourth alternative model is the base case of the ones proposed

by Ferreira and Santa Clara (2011), computed as follows:

$$(1 + r_{t+120}) = (1 + \Delta e_{t:t-240}) * (1 + dy_t)$$

where r_{t+120} is the forecast computed at time t of the future 10-year returns, $\Delta e_{t:t-240}$ is the 20-year moving average up to time t of reported earnings-per-share growth of the S&P 500 companies, dy_t is the dividend yield at time t (therefore assuming the dy follows a random walk process).

- *VAR2*. The fifth alternative model is similar to the FV-VAR2 frameworks with the difference that growth of price-to-earnings is forecasted using a frequentist VAR with no steady-state priors, whose forecasts therefore converge to the historical average of the endogenous variable.
- *VAR1*. VAR1 mirrors VAR2 apart from the assumption on dividend-price ratio that is now assumed to follow a random walk, therefore forecasting dy by using the most recent available data at each point in time.
- *VAR3*. This model mirrors the *VAR2* method apart from the assumption on earnings growth that in this case is forecasted with a frequentist VAR.

2.4 Empirical analyses - Forecast accuracy and investors economic gains

As in Cilla (2023), in order to assess the value added of the proposed framework, I compare the forecast accuracy of the proposed and alternative models in forecasting out-of-sample US equities returns. Subsequently, I also compute the economic gain of an investor that would use each of those methods to build a buy-and-hold portfolio made of US equities and US bonds with a ten-year investment horizon.

2.4.1 Data

Beyond the data used to compute the equilibrium of earnings growth presented in section 2, all data used to perform empirical analyses is downloaded from Shiller

database: earnings-per-share, dividend-per-share, price of the S&P500 index, and 10-year real yield¹¹. For all methods that use earnings, I lag earnings data by six months in order to make the analysis real-time. This also applies to the CAPE ratio that therefore is a slightly different version of the one available in Shiller database.

The full sample data used to perform empirical analyses goes from March 1947 to March 2023, with the first forecast generated in January 1970 for the 10-year period going from February 1970 to January 1980. In fact, the first 10 years of the sample are used to estimate the first fair-value data point, and 10 years are also used as estimation window for the methods that use a VAR framework. Moreover, all model performance evaluation analyses are generated using all the available data, as well as for the sub-periods with data before December 1990 and with data after December 1990, in order to analyse any potential differences in results over time. In line with Cilla (2023), the 1990 cut off date has been chosen as it represents the period in which earnings yields start to be characterised by a slower reversion to its historical average. This cut off date is also useful to isolate differences due to higher volatility of reported earnings over the last 20 years.

2.4.2 Forecast evaluation measures: RMSFE and CER

In order to assess the accuracy of each model's forecasts, I compute the out-of-sample root mean square forecast error (RMSFE), defined as:

$$RMSFE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where y_i are the realised values of the forecasted variable, $\hat{y}_i$ are the forecasts computed with the methods presented earlier, and n is the number of forecasts generated over the sample period.

Next, to assess the economic benefit for an investor that would use each method to build a portfolio with a 10-year investment horizon, I first compute the optimal

¹¹www.econ.yale.edu/shiller

portfolio made of US equities (approximated by the S&P500 index) and 10-year US Treasury bonds, assuming the investor solves the following optimisation problem in each month t :

$$w_t = \arg \min_w \mathbb{E}_t [U(W_{t+h})]$$

$$U(W_{t+h}) = \frac{W_{t+h}^{1-\gamma}}{1-\gamma}$$

$$W_{t+h} = \prod_{i=t}^{t+h} [1 + w_t * r_i^e + (1 - w_t) * r_i^b]$$

where γ is the investor risk aversion (with $\gamma \geq 0$ and $\gamma \neq 0$), W_{t+h} is the total cumulative wealth generated over h months starting from time t , h is equal to 120 (10-year of monthly data), w_t is the optimal portfolio weight to risky asset that is assumed to remain constant over the investment period, r_i^e is the realised equity return at time i , and r_i^b is the realised bond return at time i . The portfolio weights w are therefore static weights, meaning the investor rebalances the portfolio to the strategic benchmark every month until year 10. In line with Illmanen (2011) among others, I forecast future 10-year bond returns using the current 10-year nominal US Treasury yield available at the beginning of each 10-year investment period.

Subsequently, following Sakkas and Tessaromatis (2022), I analyse the economic performance of asset allocation strategies using a prediction model computing the average out-of-sample realized utility of terminal wealth of 520 asset allocation strategies defined in each month. In particular, I calculate the certainty equivalent return (CER) realized by the investor defined as:

$$CER = [(1 - \gamma)\bar{U}]^{\frac{1}{1-\gamma}} - 1$$

$$\bar{U} = \frac{1}{T_{oos}} \sum_{i=1}^{T_{oos}} \frac{W_{i+h}^{1-\gamma}}{1-\gamma}$$

where γ is the investor risk aversion, h is equal to 120, T_{oos} is the total number of months used to compute out-of-sample statistics (that in this analysis is equal to 520), W_{t+h} is the total cumulative wealth generated over h months starting

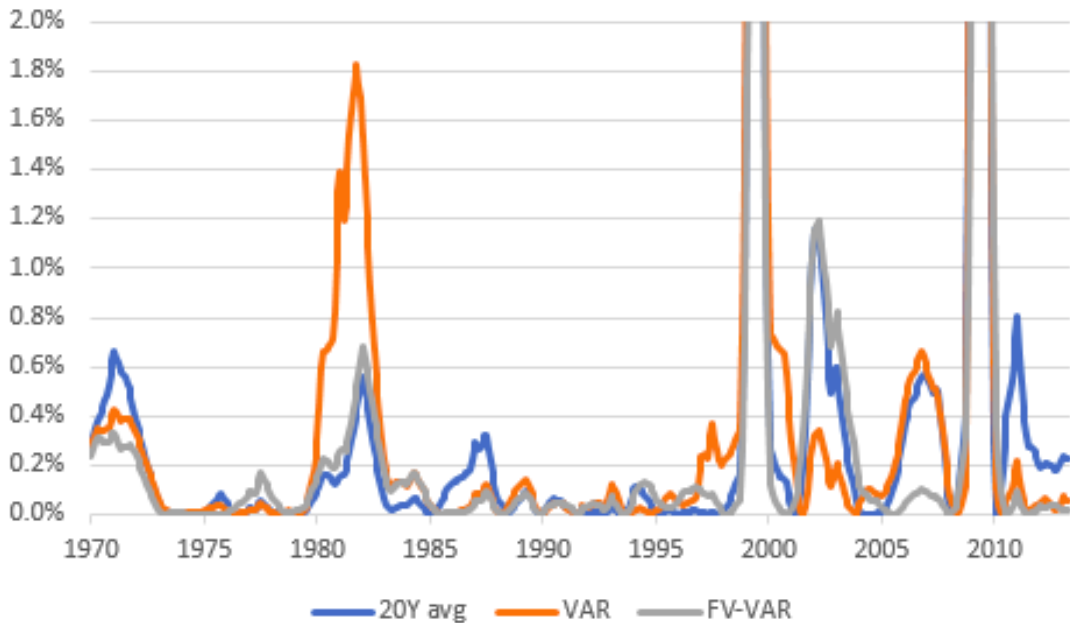
from time t .

In order to analyse the economic gains for different types of investors, I consider three cases. In the first case, I follow Ferreira and Santa Clara (2011) in setting the risk aversion γ equal to 2. In the second case, in order to create portfolios without short positions or leverage, I set the risk aversion equal to 2 and I constrain the risky asset weight w to be between 0% and 100%. In the third case, I set the risk aversion equal to 4 without constraints on w .

2.4.3 Forecast accuracy: earnings growth

To start, Figure 10 reports the time series of the out-of-sample squared forecast error of the 10-year annualised percentage earnings growth, where earnings growth is forecasted using various methods: the steady-state fair-value VAR (gray line), the 20-year moving average as in Ferreira and Santa Clara 2011 (blue line), and a standard frequentist VAR (orange line).

Figure 10: Squared forecast errors of 10-year % reported earnings growth

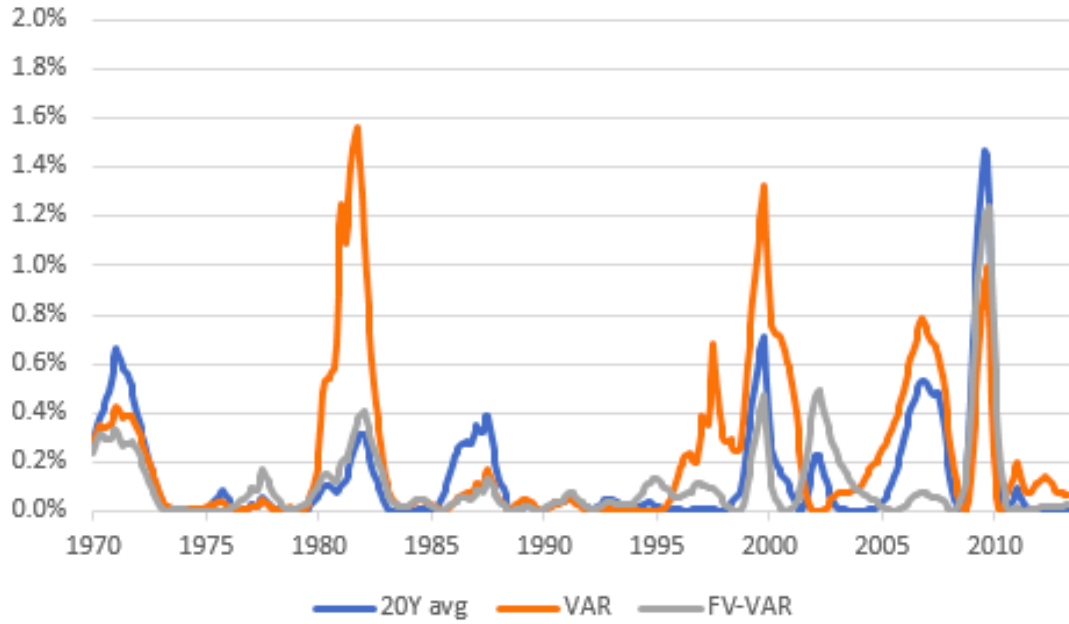


All lines are squared forecast errors (SFE) of 10-year % annualised growth rate of S&P500 index reported earnings growth computed with different methods. The blue line is the SFE of forecasts generated with a 20-year rolling average, the orange line is the SFE generated from a frequentist VAR model, the gray line is the SFE of the FV-VAR model.

Overall, the steady-state FV-VAR generates the most accurate forecasts, although the model is not always the best at any point in time. In particular, the RMSFE of the FV-VAR over the entire sample is equal to 5.8%, around one percent lower than the RMSFE of the 20-year rolling average that is equal to 6.8%. The worst performer among the models tested is the frequentist VAR with a RMSFE equal to 7.4%. In particular, the forecasts of the VAR method generated in the early 1980s were particularly inaccurate when the actual trend earnings growth reached levels close to 0% in the following 10 years. On the other hand, the simple 20-year rolling average tends to be very sensitive to the outliers that characterised earnings growth during the Global Financial Crisis, with the FV-VAR method generating more stable forecasts. For the 10-year forecasting period ending in May and June 2009, all the models were of course not able to predict the dramatic realised drop in earnings growth, however the FV-VAR model had the lowest forecasting error of 4.1%, versus 4.6% of the 20-year average and 5.6% of the VAR model. Similarly, for the forecasts generated in 2009 for the period 2009-2019, both the VAR and 20-year average method predicted a much lower value than the realised one: the FV-VAR registered a forecast error of 9.4%, while the VAR a forecast error of 17.8%, and the 20-year average a forecast error of 14.8%. It is important to stress that in all periods after the Global Financial Crisis the values of the earning growth series itself that the models are trying to forecast, is affected by a great amount of volatility. For this reason, beyond the accuracy in forecasting earnings growth, it is important to see whether these forecasts are better at ultimately forecasting returns. In fact, more stable forecasts of earnings growth, that are not necessarily the most accurate at each point in time, might contribute to generate more stable and accurate return forecasts.

As robustness check, I replicate the analyses using operating earnings that are characterised by a much lower volatility post-Global Financial Crisis.

Figure 11: Squared forecast errors of 10-year % operating earnings growth



All lines are squared forecast errors (SFE) of 10-year % annaulised growth rate of S&P500 index operating earnings growth computed with different methods. The blue line is the SFE of forecasts generated with a 20-year rolling average, the orange line is the SFE generated from a frequentist VAR model, the gray line is the SFE of the FV-VAR model.

Also in this case, the FV-VAR method performs best overall, although the gains are smaller in magnitude compared to the 20-year average method: the RMSFE of the FV-VAR is 3.3%, 40 basis points lower than that of the 20-year average method equal to 3.7%, and 140 basis points lower than that of the frequentist VAR equal to 4.7%.

2.4.4 Forecast accuracy: equity returns

Next, it is important to test if improvements in forecast accuracy of earnings growth generates more precise 10-year total return forecasts. Table 2 reports the out-of-sample RMSFE for the 10-year annualised total return of the S&P500 index using the methods described earlier. In order to test whether RMSFE of the various methods are statistically different from the RMSFE of historical average forecasts, I use a Diebold Mariano test with Newey and West (1987) standard errors with the lag truncation parameter value based on Lazarus, Lewis, Stock and Watson (2018).

Table 2: RMSFE of forecasting models

	Hist. avg.	CAPE	EY	FSC2011
All	5.9	5.24	4.75*	4.65**
pre-1990	5.7	2.51***	3.53***	3.55**
post-1990	5.96	5.89	5.09	4.97

	VAR1	VAR2	VAR3	FV-VAR1	FV-VAR2	FV-VAR3
All	4.8	4.1*	4.51	3.58**	3.49***	3.24***
pre-1990	2.05***	1.85***	1.77***	2.44**	2.15***	2.13***
post-1990	5.43	4.63	5.11	3.9**	3.84**	3.54**

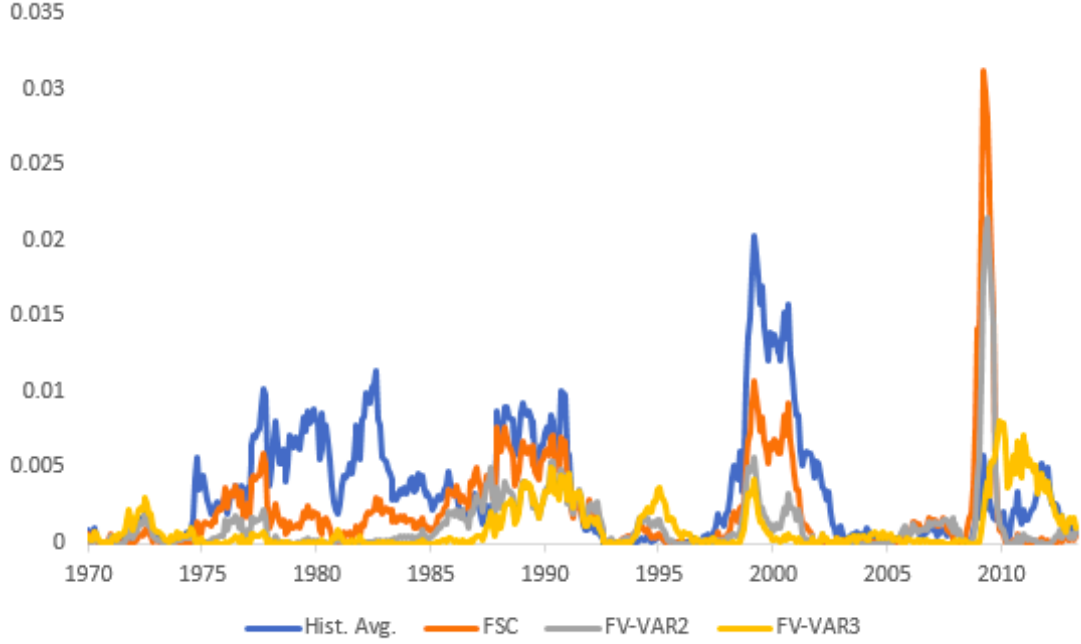
The table reports the RMSFE of each forecasting method. *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. Data used to estimate the models and to perform out-of-sample forecast evaluation is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

The results for all methods apart from the VAR3 and FV-VAR3 are the same as the ones presented in Cilla (2023). The FSC2011 method performs better than the methods based on historical average of returns, CAPE, and EY predictive regressions. CAPE performs particularly better in the periods before 1990 given the stronger mean-reversion of equity valuations. However, in the sample post-1990, none of these methods generates improvements in RMSFE that are significantly different from the simple method based on historical average. Moving on to the methods based on autoregressive models with valuations reverting to historical average, the VAR2 method reduces the RMSFE compared to the FSC2011 given a better performance particularly in the pre-1990 periods, however none of them generates statistically different results compared to the historical average method in the post-1990 sample period. Looking at VAR3 method introduced in this paper, it is interesting to see that while it is built using more accurate earnings forecasts than the VAR1 and VAR2 models, its

performance is on average better than the VAR1 but it is actually worse than the VAR2 model. This proves that the interaction among all the components that affect equity returns is important, and that improving the forecast accuracy of each component of returns does not necessarily translate into a more precise total return forecast. FV-VAR models are the best performers overall: they are the only methods that generates a reduction in RMSFE that is statistically different from the historical average method at least at 5% confidence level. They perform only slightly worse compared to the VAR frameworks only in the periods before 1990 where the reversion to the historical average of valuations tends to favour models based on this assumption, however their performance is significantly better than any other method in the post-1990 periods. In this context, the FV-VAR3 method introduced in this paper reduces the RMSFE even further compared to the other FV-VAR models: by approximately 35 basis points with respect to the FV-VAR1 model, and 25 basis points with respect to the FV-VAR2 model. The improvements are particularly visible in the post-1990 sample, characterised by a higher volatility of earnings. Lastly, this is also a confirmation of the robustness of the FV-VAR1 and FV-VAR2 models, and the importance of modelling a reversion of valuations towards an equilibrium informed by economic conditions.

Figure 12 can help to better understand the results shown in Table 2 by plotting the time series of the squared forecast errors that are used to compute the RMSFE numbers reported in Table 2. As noted in Cilla (2023), all methods that use the 20-year rolling average of earnings growth to forecast future earnings growth are particularly sensitive to the volatility of earnings generated by the periods during the Global Financial Crisis. It is now interesting to note that the FV-VAR3 method, that departs from the historical average methodology, generates much more stable forecasts in the 2009 period by significantly reducing the RMSFE over that period, although with a slightly higher error in the periods after that.

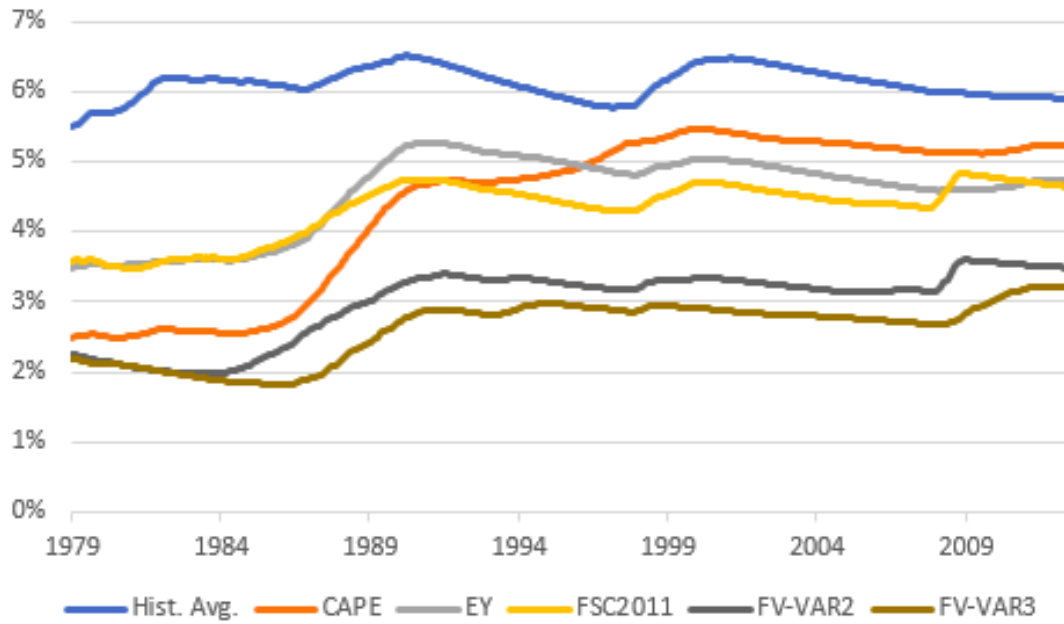
Figure 12: Squared forecast errors of 10-year US equity returns (with reported earnings)



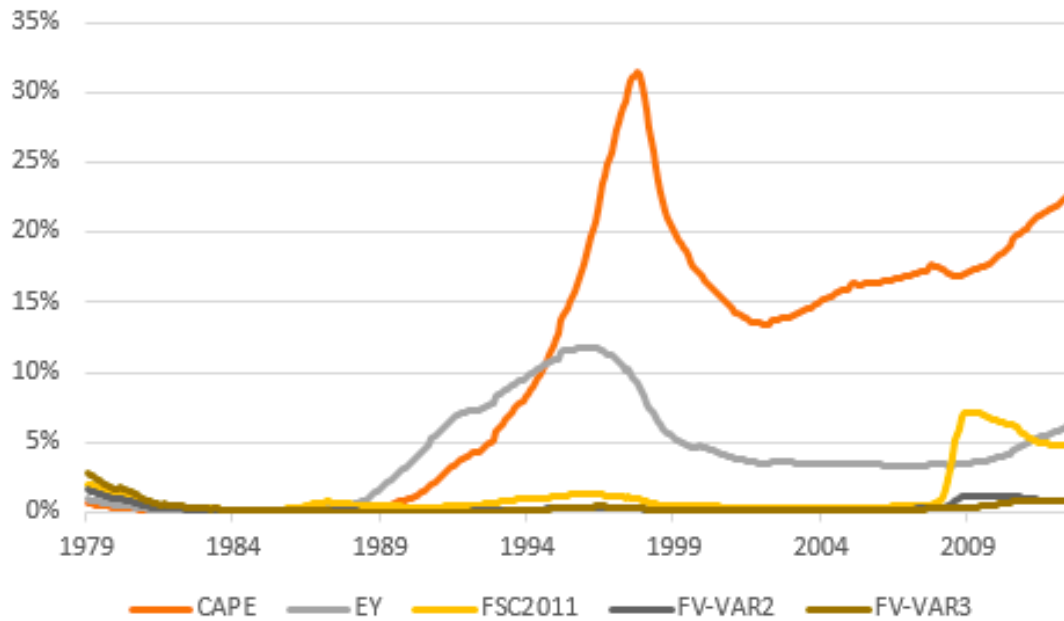
Squared forecast error of 10-year US equity returns for the following methods: historical average, FSC2011, FV-VAR2, and FV-VAR3. X-axis reports the time in which the forecasts have been generated.

As an additional analysis, inspired by Goyal and Welch (2008), Figure 13 reports the average RMSFE over rolling expanding windows, with a 10-year starting window. Therefore, the last data point of the lines reported in Figure 13 are the RMSFEs (and their p-values) reported in Table 2 for the total sample case. In addition to the results shown in Cilla (2023), here I also add the rolling RMSFE of the FV-VAR3 model, that is now the best performer among the methods tested, in all the expanding windows.

Figure 13: Rolling RMSFEs and their p-values



(a) Rolling RMSFE of forecasting models



(b) Rolling pvalues of RMSFE of forecasting models

Lines in subfigure 'a' are the RMSFE of various forecasting models, computed with rolling expanding windows with the first window equal to 10 years. Lines in subfigure 'b' are the p-values of the Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, computed with rolling expanding windows with the first window equal to 10 years. *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. X-axis reports the dates in which the forecast for the following 10-year equity returns is produced.

Figure 13b reports the p-values statistics of the RMSFEs shown in Figure 13a. In particular, these are the p-values of the Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018). Combining the results of Figure 13a and Figure 13b, we can conclude that the FV-VAR3 method produces RMSFEs that are always statistically significantly lower than the one produced by the simple historical average method. Moreover, both FV-VAR2 and FV-VAR3 produce out-of-sample results that are stable across time and always statistically significant (at least) at 5% confidence level.

As part of the robustness checks, I now test what is the impact of using operating earnings data. This has also the goal to overcome some of the problems previously mentioned about the volatility of earnings after the Global Financial Crisis. As noted by Siegel (2016), the increasing conservatism of FASB standards, especially as related to the forced write-down of asset values, has resulted in severe downward biases to reported earnings. This bias is especially severe in recessions and periods when market prices are depressed. Table 3 replicates the results reported in Table 2 with the difference that all methods that rely on earnings data are now computed with operating earnings instead of reported earnings. This means that CAPE, EY, all VAR and FV-VAR methods were re-estimated using operating earnings.

Table 3: RMSFE of forecasting models

	Hist. avg.	CAPE	EY	FSC2011
All	5.9	4.45*	4.66*	4.57**
pre-1990	5.7	2.51***	3.53***	3.55**
post-1990	5.96	4.94	4.98	4.86**

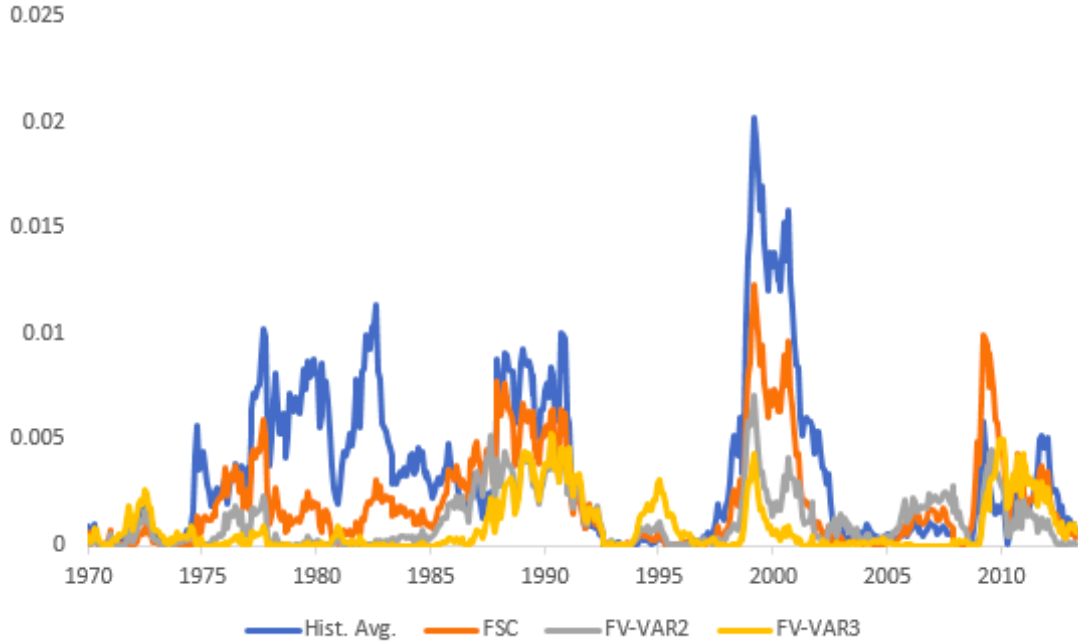
	VAR1	VAR2	VAR3	FV-VAR1	FV-VAR2	FV-VAR3
All	4.21*	3.59**	3.94**	3.51**	3.43***	2.97***
pre-1990	2.05***	1.85***	1.77***	2.44**	2.15***	2.13**
post-1990	4.73	4.01**	4.45	3.81**	3.77**	3.22***

The table reports the RMSFE of each forecasting method. *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. Data used to estimate the models and to perform out-of-sample forecast evaluation is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between RMSFE of each method versus the RMSFE of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

Looking at RMSFEs in Table 3, results are overall consistent with results generated using reported earnings. As expected, lower volatility of operating earnings with respect to reported earnings helps to decrease the RMSFE thanks to lower error of those forecasts generated using data of the period 2007–09, as shown in Figure 14. In line with Siegel (2016), the performance of CAPE model improves, with a reduction of RMSFE of around 80 basis points over the entire sample on average, and of around 1% in the post-1990 periods. As explained by Siegel (2016) this is the results of a slower mean-reversion of valuations to their historical average in the post-1990 periods, when in fact a CAPE ratio based on reported earnings would have signalled a more overvalued market than a CAPE ratio based on operating earnings. It is worth noting that using operating earnings improves the performance of the FSC2011, although just marginally by 10 basis points overall, with higher gains in the post-1990 periods when the differences from the historical average benchmark are now statistically significant. Moreover, among the methods based on autoregressive models, particularly the VAR methods tend to gain the most in terms of forecast accuracy - around 50 basis points. Improvements are more muted for the FV-VAR models, with the RMSFE of the FV-VAR3 method that decreases by around 30 basis points. At the same time, when operating earnings are used instead of reported earnings, FV-VAR3 method's RMSFE is more than 50 basis points lower than the FV-VAR1 and around 45 basis points

lower than the FV-VAR2

Figure 14: Squared forecast errors of 10-year US equity returns (with operating earnings)



Squared forecast error of 10-year US equity returns for the following methods: historical average, FSC2011, FV-VAR2, and FV-VAR3. X-axis reports the time in which the forecasts have been generated.

2.4.5 Economic gains

Next, given the improvements in the forecasting accuracy of FV-VAR model, it is now important to assess the economic benefit for an investor that would use each method to build a portfolio with a 10-year investment horizon. Table 4 reports the results in terms of CER gains of each forecasting method (in columns) compared to the historical average method, and for each type of investor (in rows). It is worth mentioning that I do not consider transaction costs or liquidity constraints as all these strategies can be implemented with low-cost and liquid ETF products.

Table 4: CER gains over historical average method (reported earnings)

	CAPE	EY	FSC2011
$\gamma = 2$	0.11	0.63	0.76*
$\gamma = 2, \text{const.}$	-0.06	0.26	0.78*
$\gamma = 4$	-0.33	0.05	-0.21*

	VAR1	VAR2	VAR3	FV-VAR1	FV-VAR2	FV-VAR3
$\gamma = 2$	-0.27	0.92*	0.52	2.4**	2.75**	2.57**
$\gamma = 2, \text{const.}$	-0.08	0.67*	0.24	1.9**	2.2***	1.85**
$\gamma = 4$	-0.93	-0.32*	-0.54	0.62***	0.75***	0.52**

The table reports the difference between certainty equivalent return (CER) of optimal portfolios built using each forecasting method indicated in columns and the CER of optimal portfolios built using the historical average forecasting method. Investor is assumed to optimise CRRA utility function with static portfolio weights and a 10-year investment horizon. Columns refer to various forecasting methods: *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. Rows refer to various investor's profiles: $\gamma=2$ is the case of an investor with risk aversion equal to 2, $\gamma=2, \text{const.}$ is the case of an investor with risk aversion equal to 2 and risky asset weight constrained between 0% and 100%, $\gamma=4$ is the case of an investor with risk aversion equal to 4. Data used is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between CER of each method versus the CER of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

Overall, the results in terms of CER tend to mirror the ones for forecast accuracy, signalling that on average more precise forecasts translate into potentially higher benefits for investors, although this is not always the case. As in Cilla (2023), among the traditional methods proposed by the literature, the FSC2011 performs best for more risk-seeker investors, with gains of around 80 basis points. For a more risk-averse investor, with γ equal to 4, both CAPE and FSC2011 actually have negative utility gains, while EY method generated gains just above the historical average method. For VAR methods, while results are more positive for a risk-seeker investor that would use a VAR2 model, the VAR3 introduced in this paper does not add utility gains, in line with the fact that its return forecasts are not more precise than other VAR methods. Moving on to the FV-VAR methods, while the FV-VAR3 generates more precise forecasts overall,

those gains translate into marginally higher utility compared to the FV-VAR1 method, and slightly lower compared to the FV-VAR2 method. These results are valid for an investor with any risk profile. Overall, FV-VAR models are still the best performers among all frameworks tested.

Moving on to the analysis of CER for the case of operating earnings, also in this case improvements in RMSFE translate into increases in investors' utility as shown in Table 5. Analysing the effect of using operating earnings and looking at the difference of the results presented in Table 5 with respect to Table 4, improvements are more significant for a risk-seeking investor (risk aversion equal to two in our case) for which the full-sample CER gain goes up by more than 1% for the CAPE method compared to the case of reported earnings, by 30 basis points for EY method, and 45 bps for FSC2011 method. Looking at the VAR methods, the increase in CER is almost 1.2% per annum compared to the case of reported earnings, while for the FV-VAR methods is around 50 basis with these methods still best performers in absolute terms among all the models we tested.

Table 5: CER gains over historical average method (operating earnings)

	CAPE	EY	FSC2011
$\gamma = 2$	1.23*	0.91	1.22*
$\gamma = 2, \text{const.}$	0.64	0.54	0.81
$\gamma = 4$	0.3	0.24	0.31*

	VAR1	VAR2	VAR3	FV-VAR1	FV-VAR2	FV-VAR3
$\gamma = 2$	0.93	2.13**	1.47	2.84***	3.34***	3.11***
$\gamma = 2, \text{const.}$	0.3	1.38*	0.86	2.24***	2.74***	2.38***
$\gamma = 4$	-0.02	0.6**	0.02*	1.03***	1.26***	0.88**

The table reports the difference between certainty equivalent return (CER) of optimal portfolios built using each forecasting method indicated in columns and the CER of optimal portfolios built using the historical average forecasting method. Investor is assumed to optimise CRRA utility function with static portfolio weights and a 10-year investment horizon. Columns refer to various forecasting methods: *Hist. Avg.* is the method based on simple historical average of realised returns, *CAPE* is the method based on the predictive regression using CAPE ratio, *EY* is the method based on the predictive regression using $1/\text{CAPE}$, *FSC2011* is the Ferreira and Santa Clara (2011) method, *VAR* methods are versions of the FSC2011 with mean reverting valuations estimated with a frequentist VAR, *FV-VAR* methods are versions of the FSC2011 with valuations reverting to a fair-value based on real yields, macroeconomic volatility and earnings growth estimated with a steady-state Bayesian VAR. Rows refer to various investor's profiles: $\gamma=2$ is the case of an investor with risk aversion equal to 2, $\gamma=2, \text{const.}$ is the case of an investor with risk aversion equal to 2 and risky asset weight constrained between 0% and 100%, $\gamma=4$ is the case of an investor with risk aversion equal to 2. Data used is from March 1947 to March 2023. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels of Diebold and Mariano (1995) test statistics computed on the difference between CER of each method versus the CER of the historical average method, with standard errors corrected using the Newey and West (1987) methodology and the value of the lag truncation parameter calculated using the formula recommended in Lazarus, Lewis, Stock and Watson (2018).

2.5 Conclusions

In this paper I extended the framework presented in Cilla (2023) to analyse various assumptions for the earnings growth component of US equity returns. To start, I tested the use of various earnings measures - reported earnings and operating earnings of the S&P500 index. In order to forecast earnings growth, I tested three models: the 20-year moving average proposed by Ferreira and Santa Clara (2011) and used in Cilla (2023), an autoregressive model whose long-term forecasts converge to the historical average, and a newly introduced autoregressive model whose long-term forecasts converge to an equilibrium informed by economic growth. For the last option, I adapt a steady-state Bayesian VAR introduced by Villani (2009). Compared to the alternatives, the proposed methodology FV-VAR3 improves the out-of-sample forecast accuracy of long-term earnings growth and contributes to generate more stable forecasts compared to alternative methods, despite the higher volatility of earnings in the years post-2009. Moreover, the proposed method helps to reduce the out-of-sample root mean square forecast error of US equity returns over an horizon of 10 years: by further 35 basis points

annualised with respect to the FV-VAR1 model and 25 basis point with respect to the FV-VAR2 model, introduced in Cilla (2023). The gains are particularly visible in the post-1990 sample characterised by higher earning volatility. Interestingly, when operating earnings are used instead of reported earnings, FV-VAR3 method's RMSFE is more than 50 basis points lower than the FV-VAR1 and around 45 basis points lower than the FV-VAR2. Finally, the FV-VAR3 method also helps to generate utility gains for investors that overall are in line with the FV-VAR frameworks: slightly higher than the FV-VAR1 and slightly lower than the FV-VAR2.

2.6 References

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